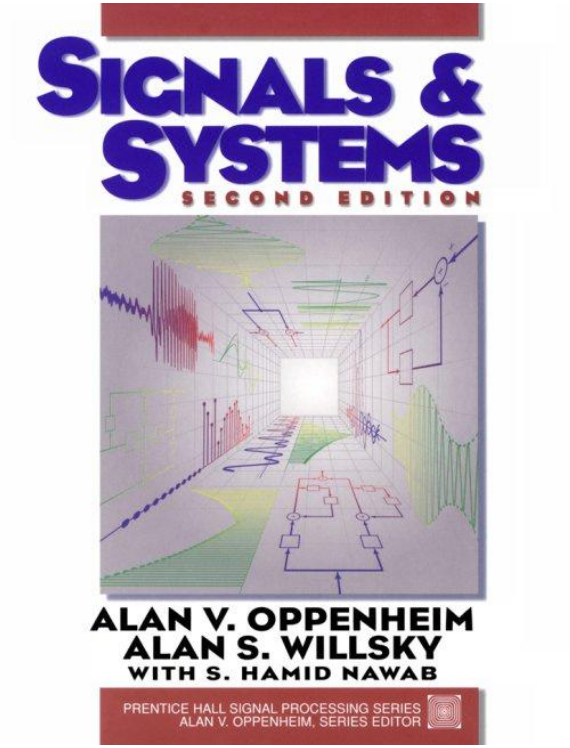
Signals and Systems - A Review

Southern University of Science and Technology

Fall 2021

Scope of Lecture

- "Signals and Systems", Oppenheim, Willsky and Nawab, 2nd Edition, 1997, Prentice-Hall.



- This course teaches Chapters 1 to 8.
  - Roughly two weeks for one chapter
  - Middle-term exam for Chapters 1 to 4
  - Final exam for all

Textbook reading is crucial, as I cannot cover every detail in slides

Basic knowledge review

- Basic knowledge on signal computation (Reverse, time shift, etc.)
- Exponential signals: Euler's relation, periodic
- CT/DT unit impulse/step function
- System Properties
  1. Memoryless or with memory
  2. Causality
  3. Invertibility
  4. Stability
  5. Time-invariance
  6. Linearity

Basic knowledge review

- CT/DT LTI systems
- Convolution operation procedure
  1. Figure computation based on “Flip-slide-multiply-sum/integral”
  2. Some known or typical convolution results
  3. Properties of convolution
- Unit impulse response and properties of LTI systems
  1. Memoryless or with memory
  2. Causality
  3. Invertibility
  4. Stability

Basic knowledge review

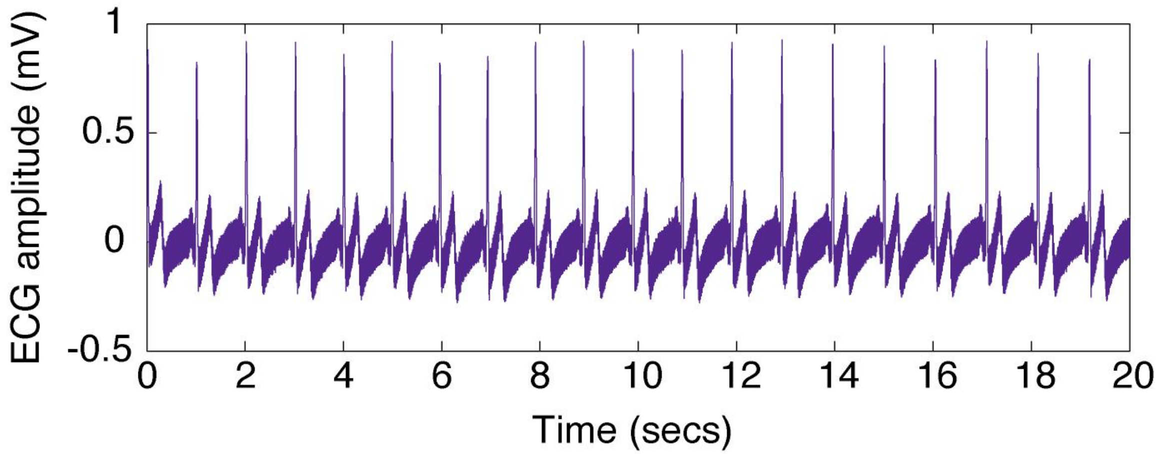
- Eigenvalue & Eigenfunction
- CT/DT Fourier Series & Properties
- CT/DT Fourier Transform & Properties
    1. Periodicity, continuity, duality
    2. Convolution/Multiply and Fourier Transform
- Differential/difference equations & frequency response
- Filter characteristics
- Sinusoidal modulation & demodulation
- Nyquist sampling rule

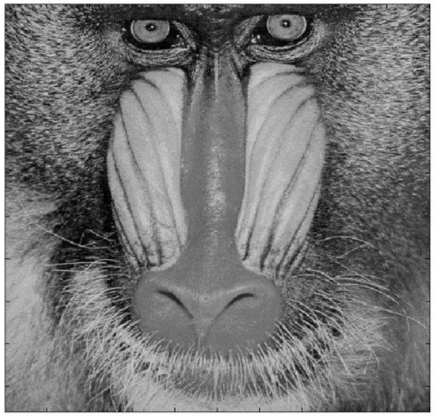
Signals and Systems

- **Signal:** everything which carries information
- **System:** everything which processes input signal and generate output signal

Signal Classification 1: Dimension of Independent Variable

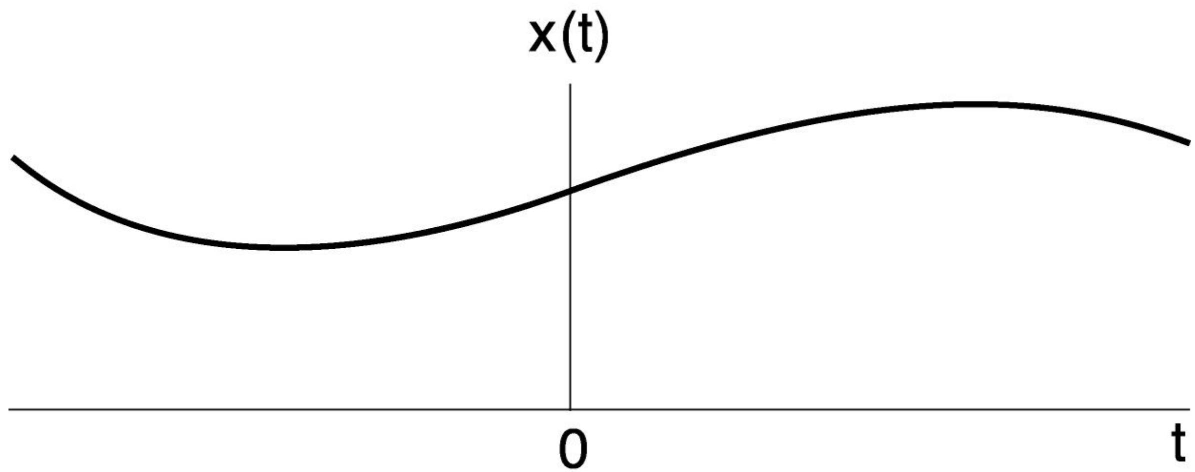
- An independent variable can be 1-D (t in the ECG), 2-D (x, y in an image), or 3-D (x, y, t in an video).





- We focus on 1-D for mathematical simplicity but the results can be extended to 2-D or even higher dimensions.

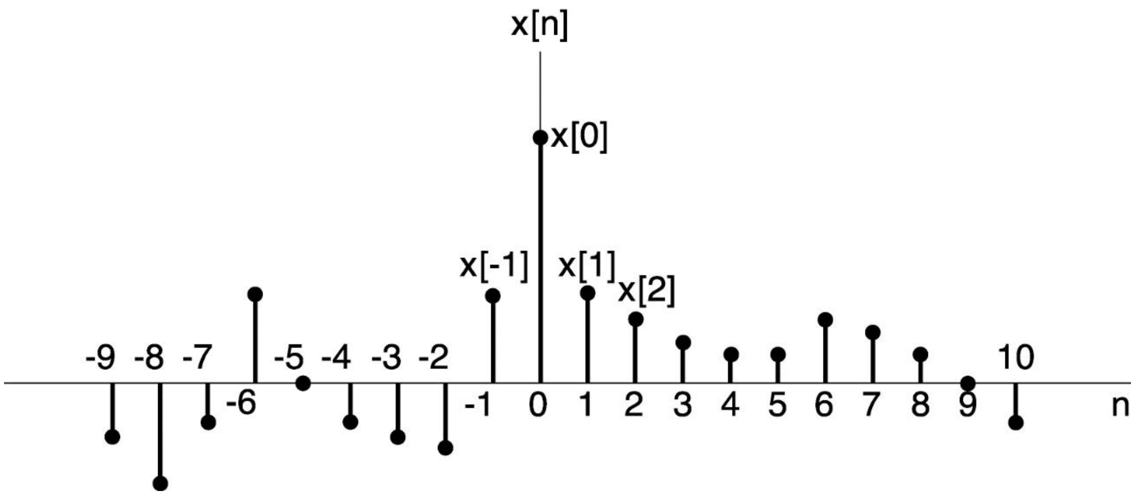
Signal Classification 2: CT/DT
Continuous-time (CT) Signals



- Independent variable is continuous
- Most of the signals in the physical world are CT signals.
- E.g. voltage & current, pressure, temperature, velocity, etc.

Notation: x(t)

Discrete-time (DT) Signals



- Independent variable is integer
- Examples of DT signals: DNA sequence, population of the *n*-th generation of certain species

Notation: $x[n]$

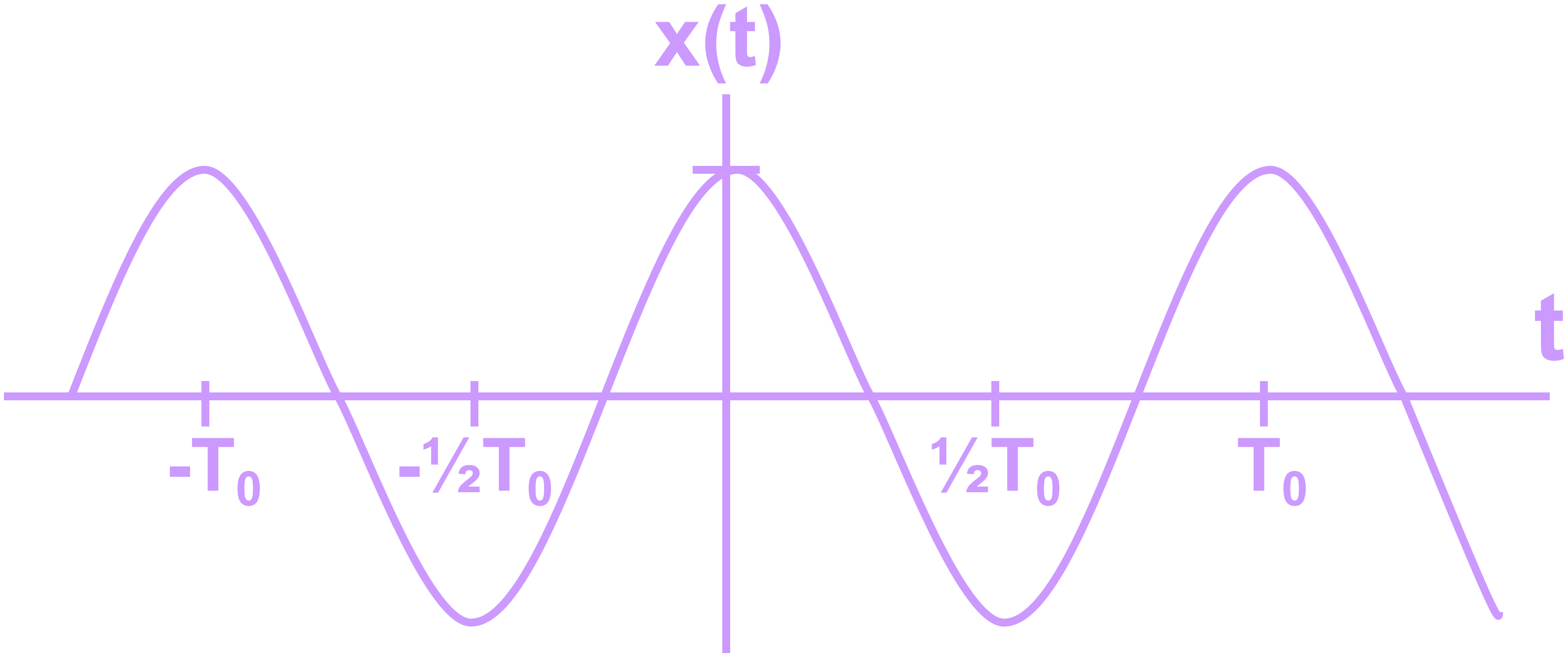
Many Human-made Signals are DT

Weekly Dow-Jones industrial average

Digital image

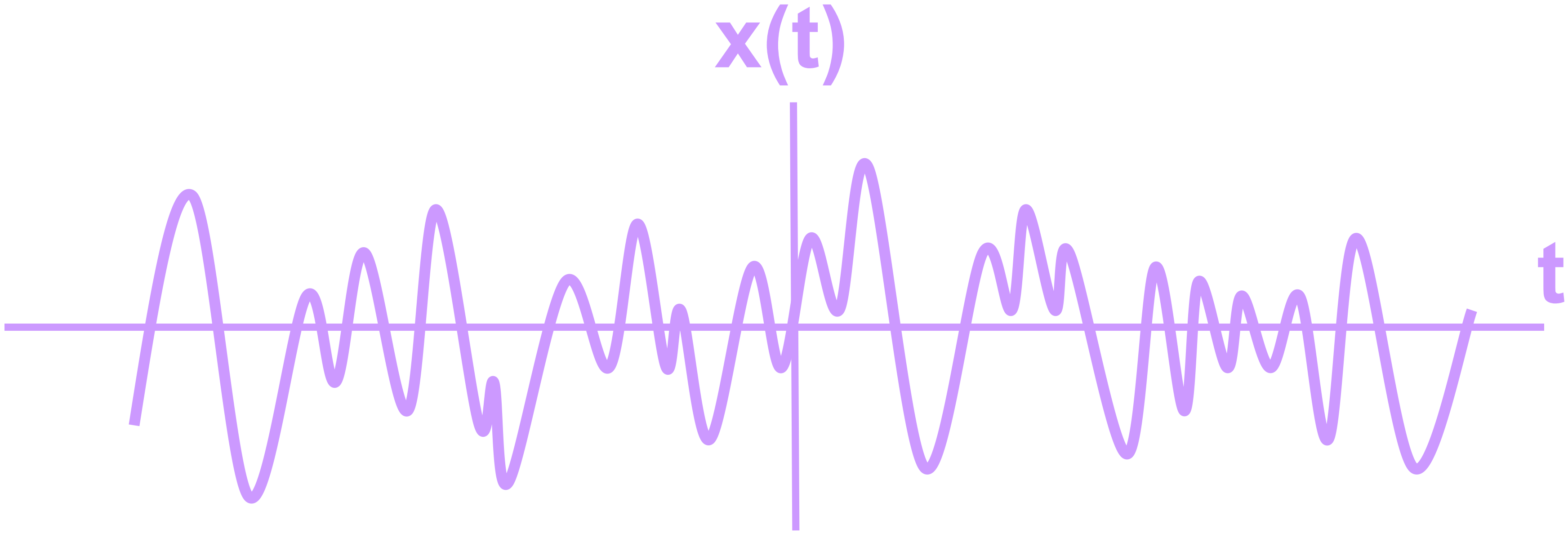
- Why DT? — Can be processed by modern digital computers and digital signal processors (DSPs).

Signal Classification 3: Deterministic /Random
Deterministic Signal



- Each value of the signal is fixed, and can be determined by a mathematical expression, rule, or table.

Signal Classification 3: Random Signal



- Signal value at any time instance is a random variable.

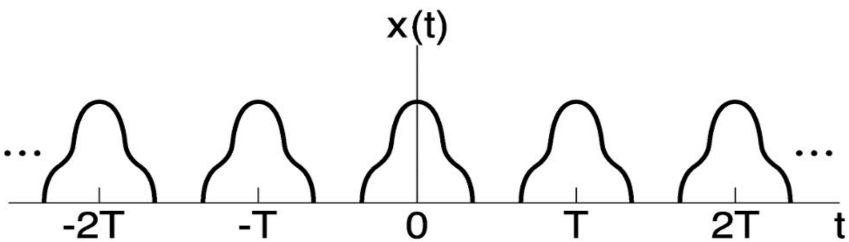
Signal Classification 4: Periodic / Aperiodic

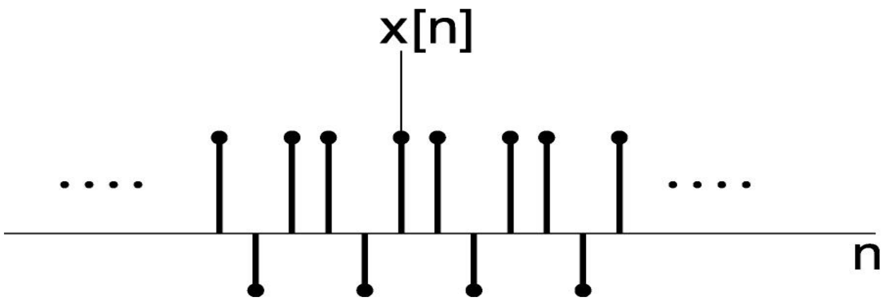
- Periodic Signals

  CT: $x(t) = x(t + T)$, T : *period*

  $x(t) = x(t + mT)$, m: *integer*

  DT: $x[n] = x[n + N]$=$x[n + mN]$, N: period





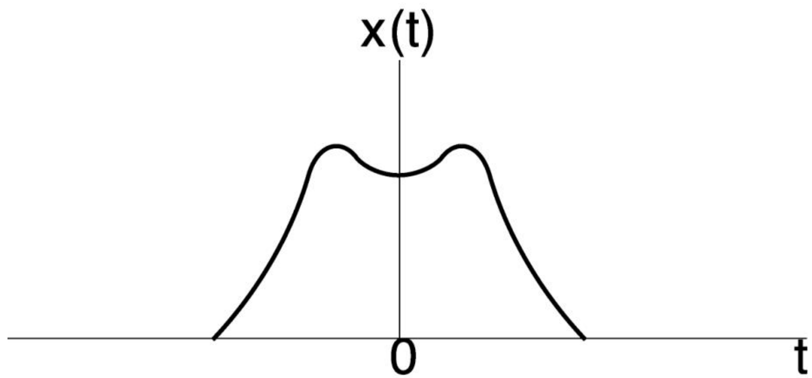
- Fundamental period: the smallest positive period

- Aperiodic: NOT periodic

Signal Classification 5: Even / Odd

- Even and Odd Signals

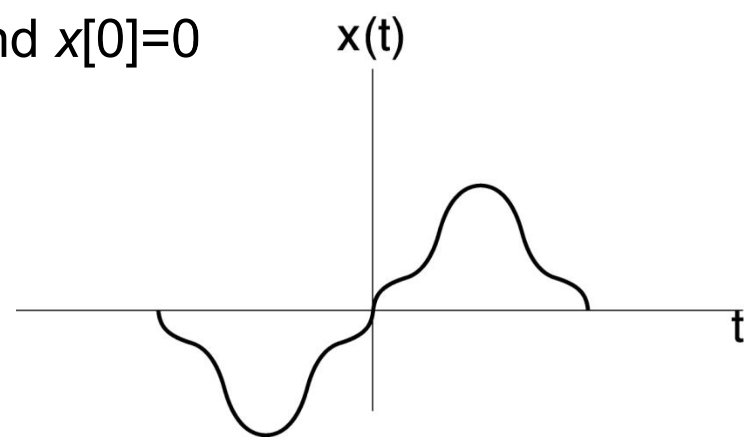
Even $x(t) = x(-t)$ *or* $x[n] = x[-n]$



Example: cos(t)

Odd $x(t) = -\,x(-t)$ *or* $x[n] = -\,x[-n]$

$x(0)=0$, and $x[0]=0$



Example: sin(t)

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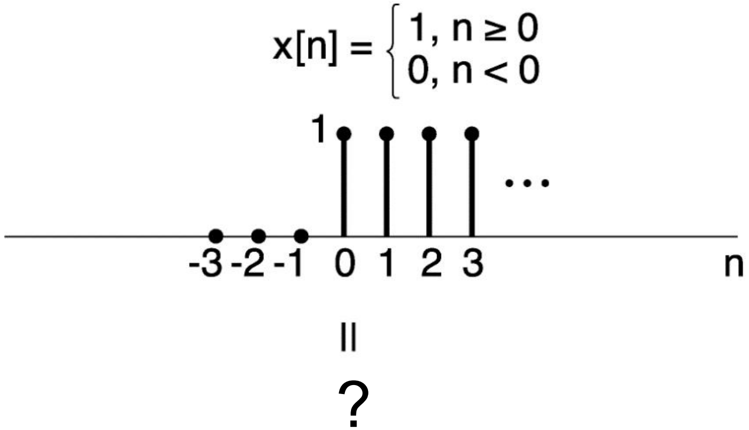
- Any signals can be expressed as a sum of *Even* and *Odd* signals. That is:

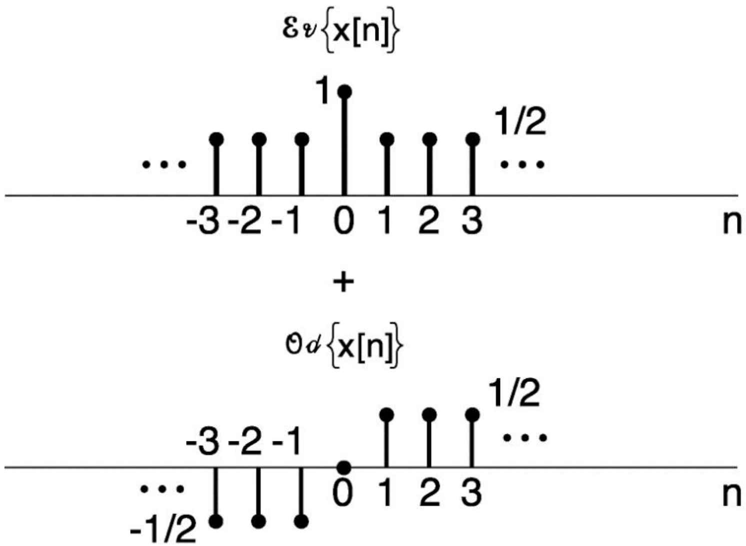
$x(t) = x_{even}(t) + x_{odd}(t)$,

where:

$x_{even}(t) = [x(t) + x(-t)]/2$,

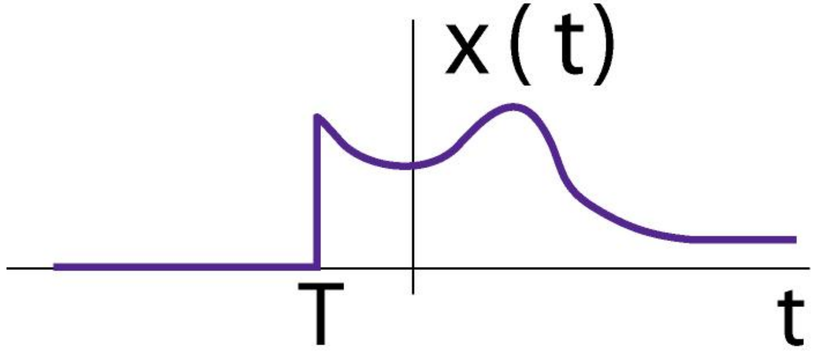
$x_{odd}(t) = [x(t) - x(-t)]/2$.

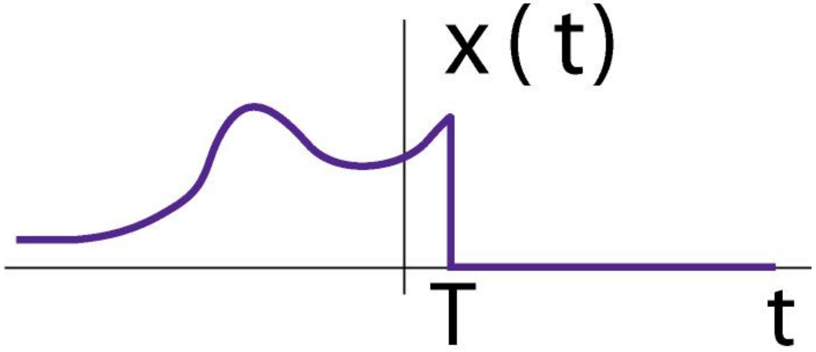




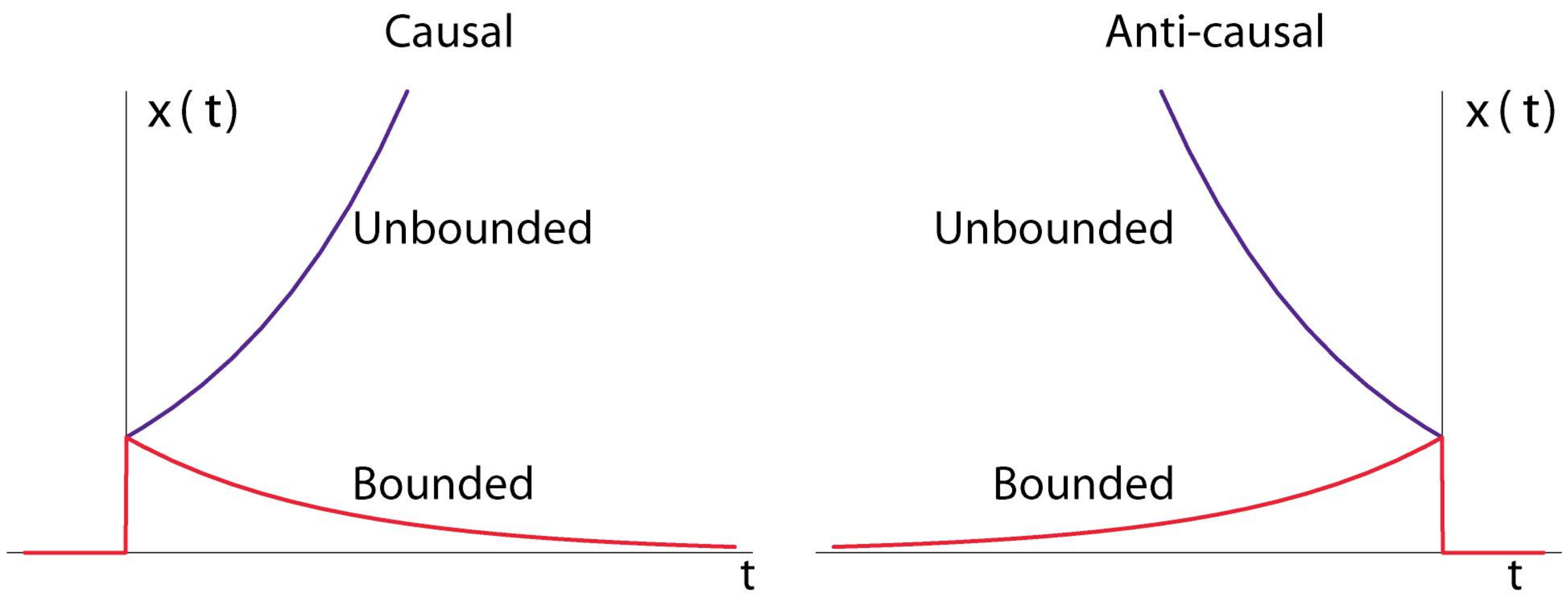
Signal Classification 6: Right- and Left-Sided

- A right-sided signal is zero for $t < T$, and
- A left-sided signal is zero for $t > T$, where T can be positive or negative.





Classification 7: Bounded and Unbounded



- Bounded signal: the absolute value of signal is bounded.
- Unbounded signal: otherwise

$$\exists C,\ |x(t)| \leq C\ \forall t$$

Transformation of a Signal

- Time Shift

$$x(t) \to x(t-t_0) \quad , \quad x[n] \to x[n-n_0]$$

- Time Reversal

$$x(t) \to x(-t) \quad , \quad x[n] \to x[-n]$$

- Time Scaling

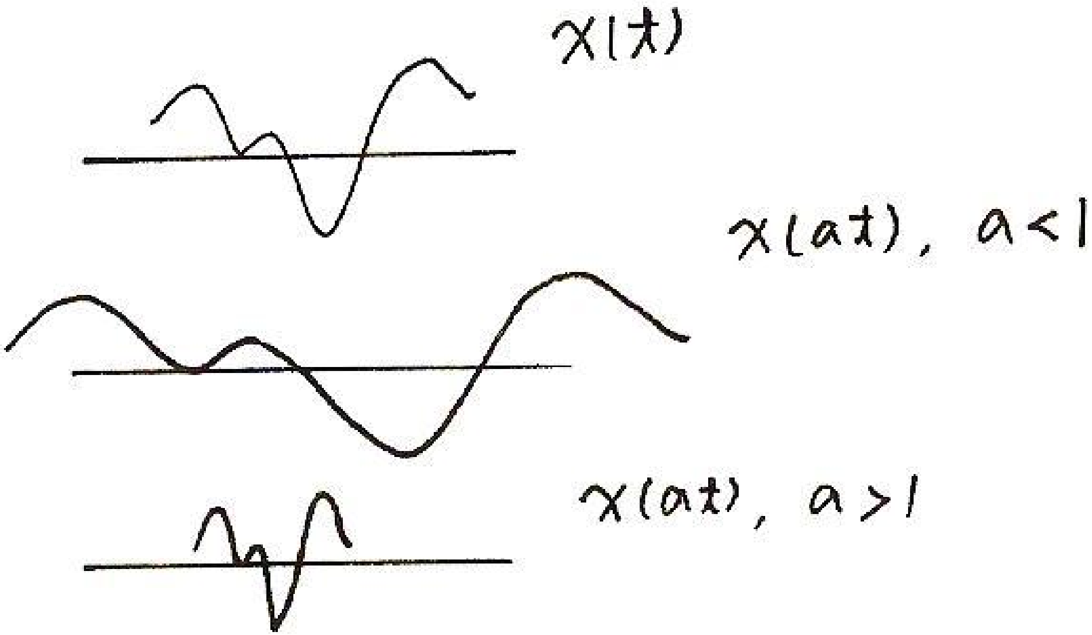
$$x(t) \to x(at) \quad , \quad x[n] \to ?$$

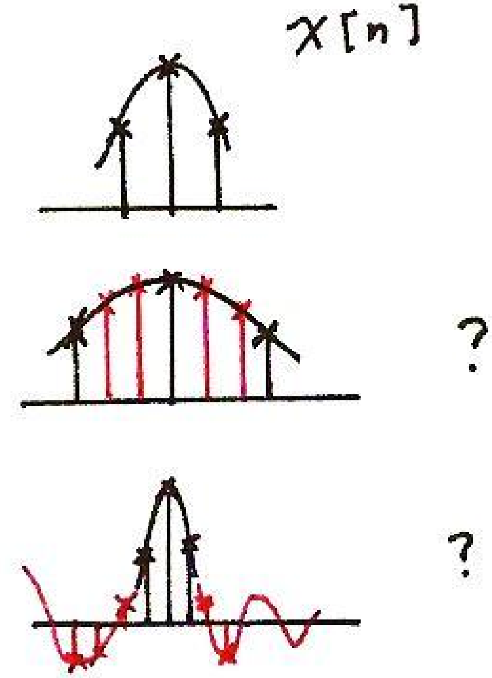
- Combination

$$x(t) \to x(at+b) \quad , \quad x[n] \to ?$$

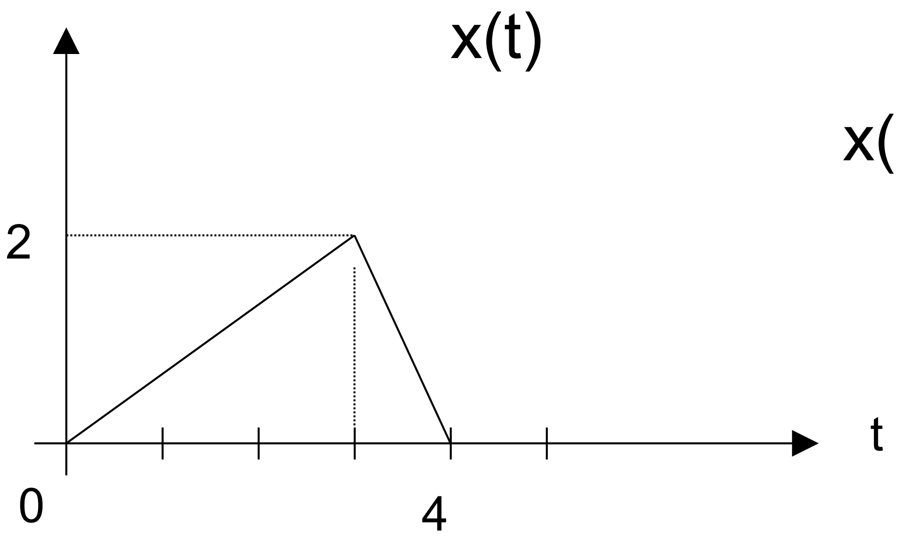
Transformation of a Signal

Time Scaling

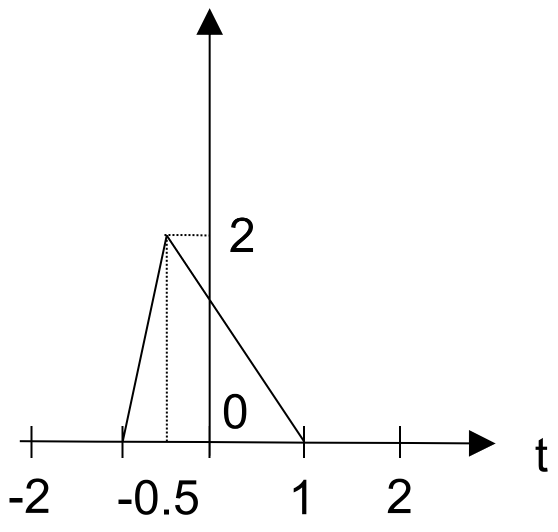




Class problem



x(-2t+2) ?

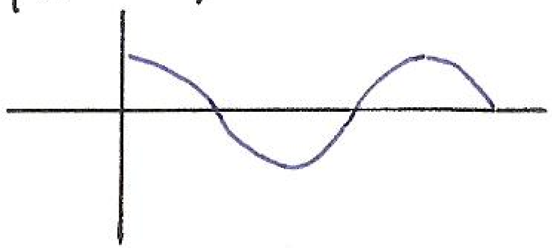


Exponential Signals

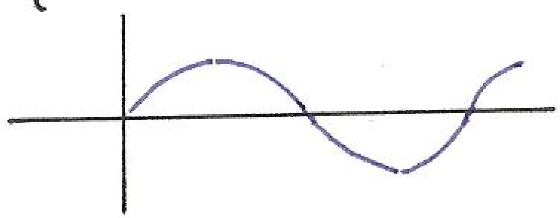
- A very important class of signals is presented as:
  - CT signals of the form $x(t) = e^{j\omega t}$
  - DT signals of the form $x[n] = e^{j\omega n}$
- For both *exponential* CT and DT signals, x is a complex quantity and has:
  - a real and imaginary part [i.e., *Cartesian form*], or equivalently
  - a magnitude and a phase angle [i.e., *polar form*].
- We will use whichever form that is convenient.

$$x(t) = e^{j\omega_0 t} = \cos(\omega_0 t) + j\sin(\omega_0 t)$$

$Re\{e^{j\omega_0 t}\} = \cos\omega_0 t$



$Im\{e^{j\omega_0 t}\} = \sin\omega_0 t$



-Fundamental (angular) frequency: $|\omega_0|$

-Fundamental period: $T_0 = \dfrac{2\pi}{|\omega_0|}$

-In CT, $e^{j\omega_0 t}$ **always periodic**

-larger ω_0 **=> higher frequency**

$$x[n] = e^{j\omega_0 n} = cos\omega_0 n + \mathrm{j}\ \mathrm{sin}\omega_0 n$$

Is it periodic?
Larger ω_0 => higher frequency?

$$e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

$$e^{j2\pi n} = (e^{j2\pi})^n = (1)^n = 1$$

$$x[n] = e^{j\omega_0 n} = cos\omega_0 n + \mathrm{j}\ \mathrm{sin}\omega_0 n$$

Is it periodic?
Larger ω_0 => higher frequency?

$$e^{j\omega_0 n} = e^{j\omega_0 (n+N)} = e^{j\omega_0 n}$$

$$\Rightarrow e^{j\omega_0 N} = 1 \Rightarrow \omega_0 N = 2\pi m \Rightarrow \frac{\omega_0}{2\pi} = \frac{m}{N}$$

Periodicity Properties of DT Complex Exponentials

Important difference between $e^{j\omega_0 n}$ and $e^{j\omega_0 t}$:

- $e^{j\omega_0 n}$ is periodic w.r.t. ω_0

$$\boxed{e^{j(\omega_0+m\cdot 2\pi)n}=e^{j\omega_0 n}\cdot e^{jm\cdot 2\pi n}=e^{j\omega_0 n}}$$

- However, $e^{j\omega_0 t}$ *is aperiodic w.r.t.* ω_0

$$\boxed{\forall x\neq 0,\ e^{j(\omega_0+x)t}=e^{j\omega_0 t}e^{jxt}\neq e^{j\omega_0 t}}$$

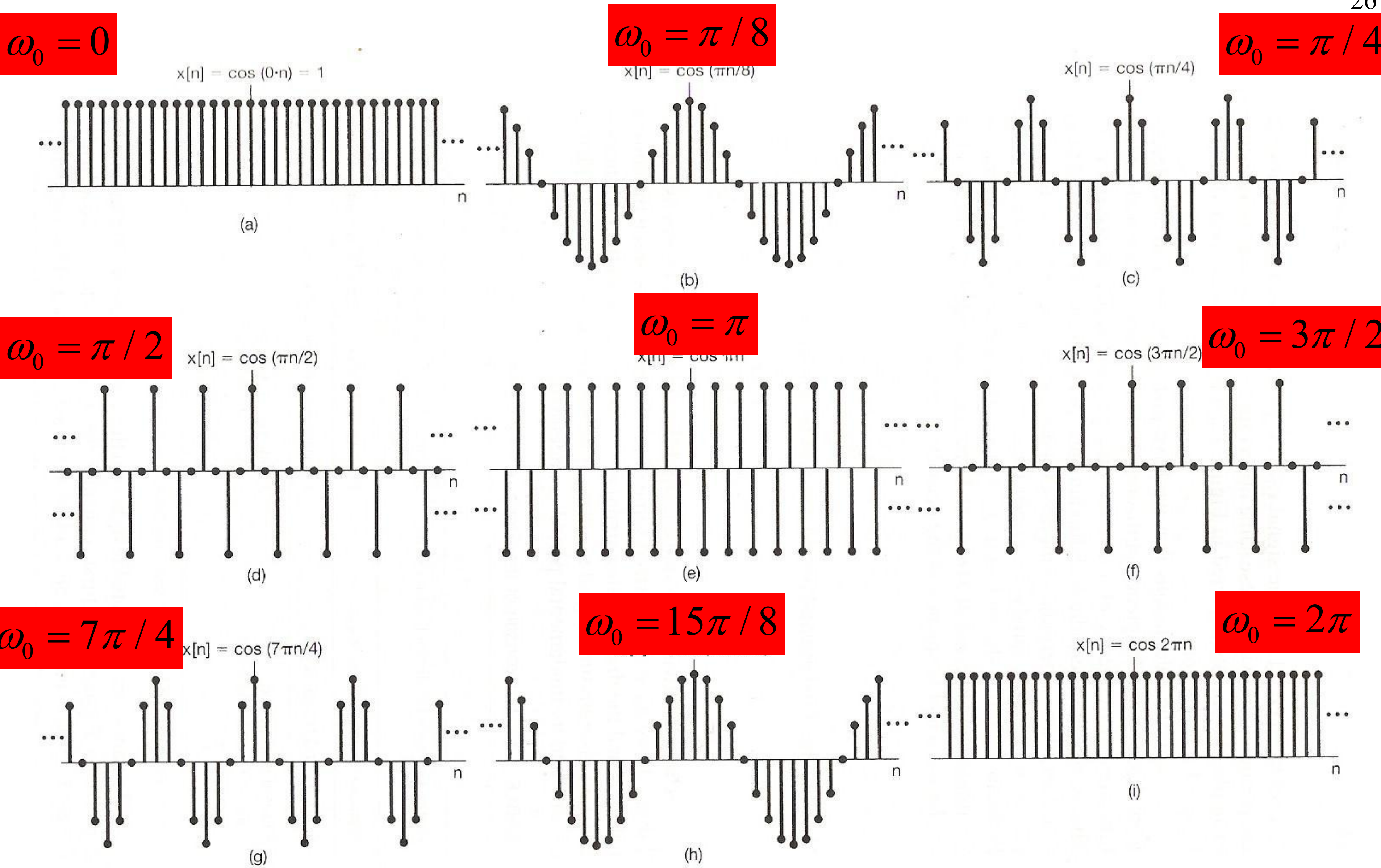


Figure 1.27 Discrete-time sinusoidal sequences for several different frequencies.

- We need only consider a frequency interval of length 2π, and on most cases, we use the interval: $0 \le \omega_0 < 2\pi$, or $-\pi \le \omega_0 < \pi$

- $e^{j\omega_0 n}$ does **not** have a continually increasing rate of oscillation as ω_0 is increased.

  lowest-frequency (slowly varying): ω_0 near 0, 2π, …, or 2k • π

  highest-frequency (rapid variation): ω_0 near ± π, …, or (2k+1) • π

$$e^{j(2k+1)\pi n} = e^{j\pi n} = (e^{j\pi})^n = (-1)^n$$

$$e^{j2\pi n} = (e^{j2\pi})^n = (1)^n = 1$$

Harmonically Related Signal Sets

- A set of periodic exponentials which have a common period.

$$\{\phi_k(t) = e^{jk\omega_0 t}, k = 0, \pm 1, \pm 2, \ldots\}$$

Fundamental (Angular) Frequency : $|k\omega_0|$

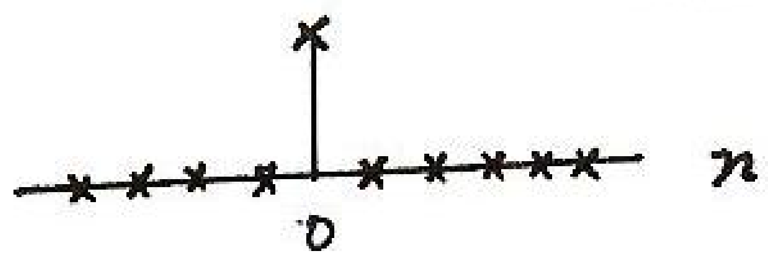
Fundamental Period: $\dfrac{2\pi}{|k\omega_0|}$

Common Period: $\dfrac{2\pi}{|\omega_0|}$

Unit Impulse (or Unit Sample) Function

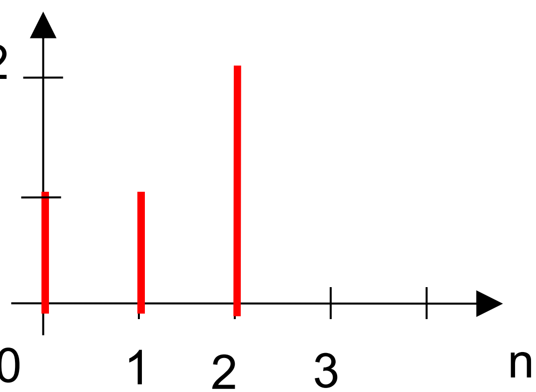
Discrete-time

$$\delta[n]=\begin{cases}0, & n\neq 0\\ 1, & n=0\end{cases}$$



- As a basic building function, we can use unit impulse function to represent other different signals.

e.g.1



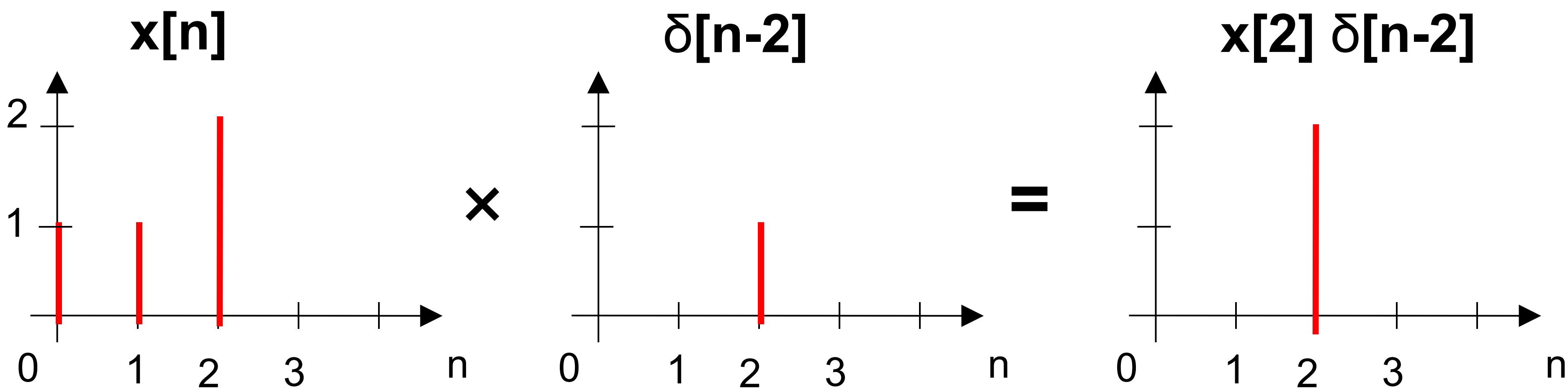
$$=\delta[n]+\delta[n-1]+2\delta[n-2]$$

Unit Impulse Function (cont.)

- Sampling property

  $x[n]\ \delta[n] = x[0]\ \delta[n]$

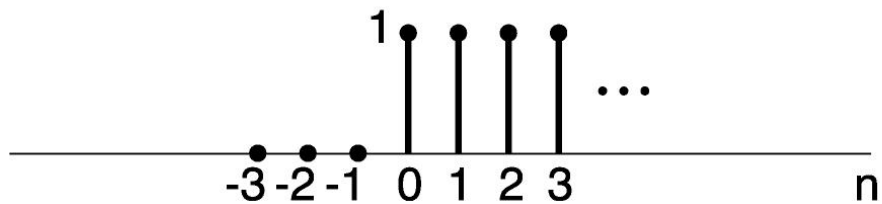
  $x[n]\ \delta[n-n_0] = x[n_0]\ \delta[n-n_0]$



Unit Step Function

Discrete-time

$$x[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



Relation between unit impulse and unit step functions

– First difference

$$\delta[n] = u[n] - u[n-1]$$

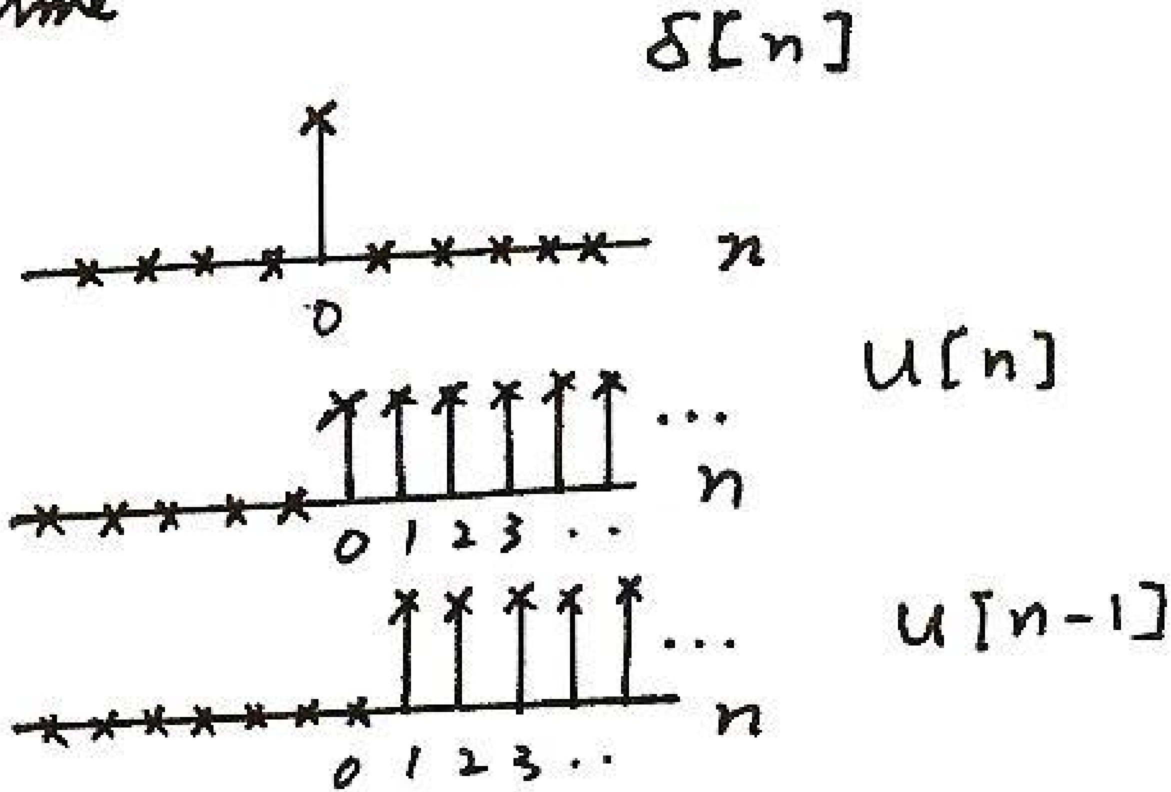
– Running Sum

$$u[n] = \sum_{m=-\infty}^{n} \delta[m] \begin{cases} =0, & n<0 \\ =1, & n \geq 0 \end{cases}$$

$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$

Unit Step Function: First Difference

Discrete-time



$$\delta[n] = u[n] - u[n-1]$$

Unit Step Function: Running Sum

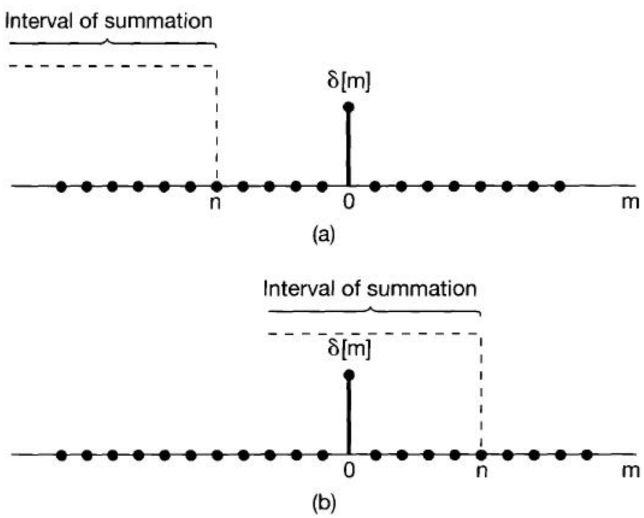


Figure 1.30 Running sum of eq. (1.66): (a) $n < 0$; (b) $n > 0$.

$$u[n] = \sum_{m=-\infty}^{n} \delta[m].$$

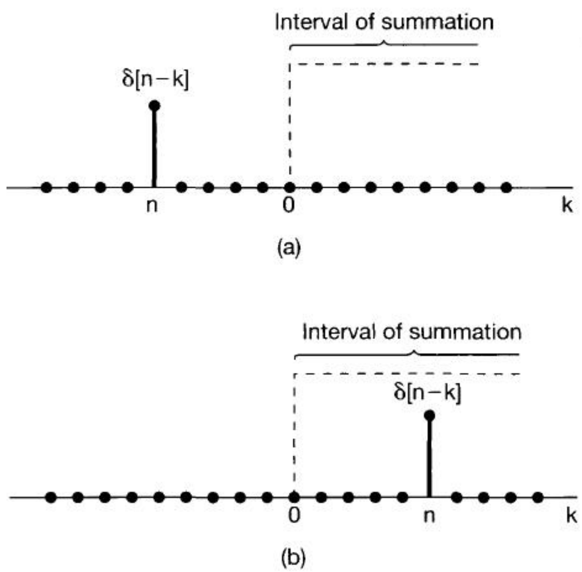
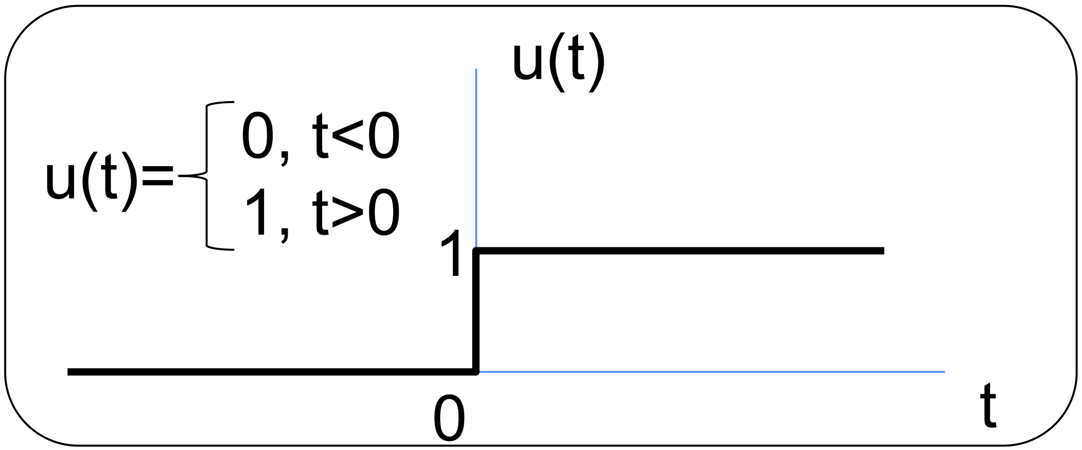


Figure 1.31 Relationship given in eq. (1.67): (a) $n < 0$; (b) $n > 0$.

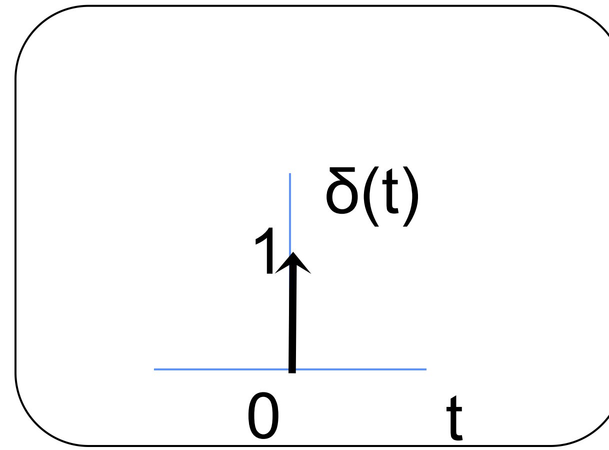
$$u[n] = \sum_{k=0}^{\infty} \delta[n-k].$$

Unit Impulse and Unit Step Functions

Continuous-time



$$u(t)=\begin{cases}0, & t<0\\ 1, & t>0\end{cases}$$



Relation between unit impulse and unit step functions

– First Derivative

$$\delta(t)=\frac{du(t)}{dt}$$

– Running Integral

$$u(t)=\int_{-\infty}^{t}\delta(\tau)d\tau$$

Unit Impulse and Unit Step Functions (Cont.)

$$\delta_\Delta(t) = \frac{du_\Delta(t)}{dt}$$

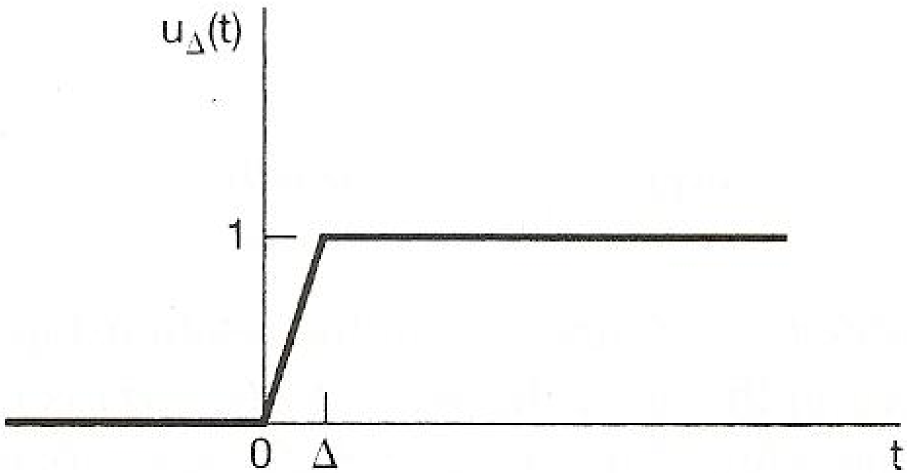


Figure 1.33 Continuous approximation to the unit step, $u_\Delta(t)$.

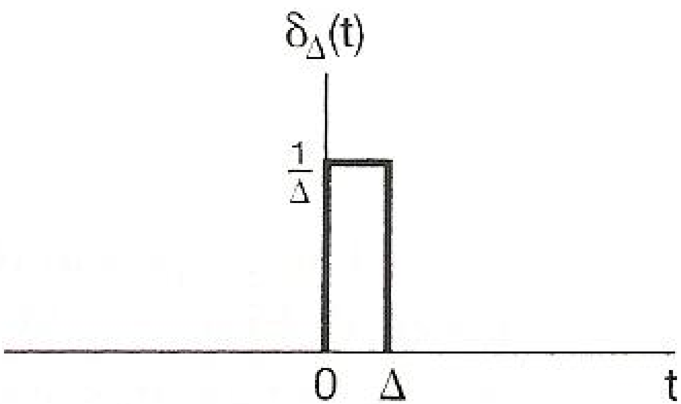


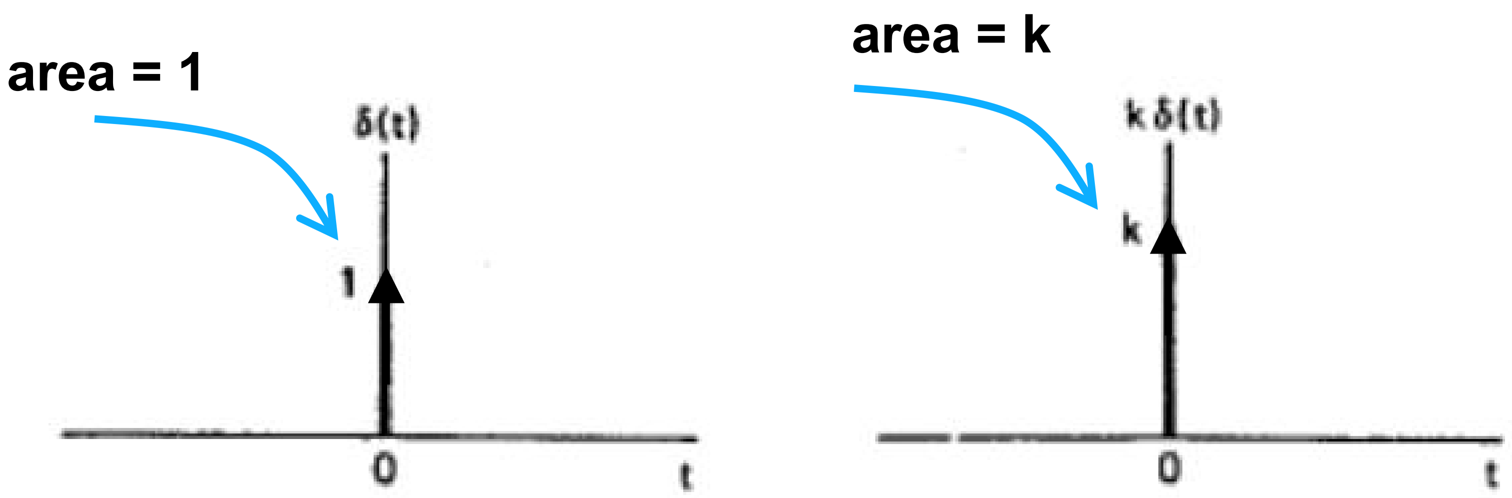
Figure 1.34 Derivative of $u_\Delta(t)$.

$$u(t) = \lim_{\Delta \to 0} u_\Delta(t)$$

$$\delta(t) = \lim_{\Delta \to 0} \delta_\Delta(t)$$

More on CT unit impulse function:

- **δ(t) has in effect no duration, but unit area.**



CT unit impulse

Scaled impulse

- **Or the integration of CT unit impulse function is unit.** $\int_{-\infty}^{\infty} \delta(t)dt = 1$

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More on CT unit impulse function:

Sampling Property:

$$x(t)\delta(t) = x(0)\delta(t).$$

$$x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0).$$

Running Integral:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau = \int_{\infty}^{0} \delta(t - \sigma)(-d\sigma),$$

$$\sigma = t - \tau$$

$$u(t) = \int_{0}^{\infty} \delta(t - \sigma)\,d\sigma.$$

More on CT unit impulse function:

Running Integral:

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau = \int_{\infty}^{0} \delta(t-\sigma)(-d\sigma),$$

$$\sigma = t - \tau$$

$$u(t) = \int_{0}^{\infty} \delta(t-\sigma)\, d\sigma.$$

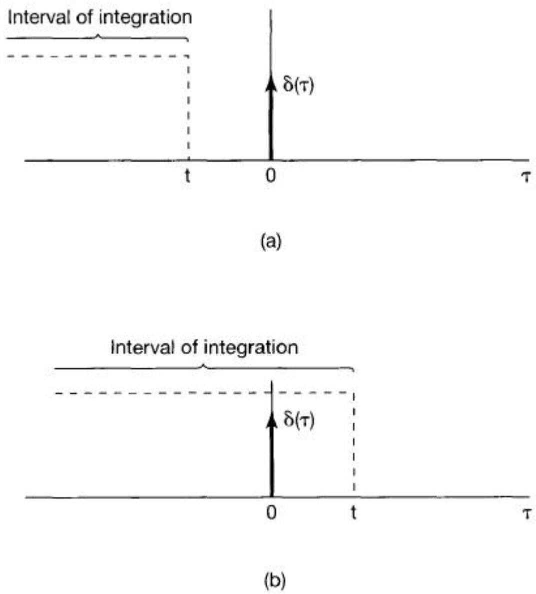


Figure 1.37 Running integral given in eq. (1.71): (a) $t < 0$; (b) $t > 0$.

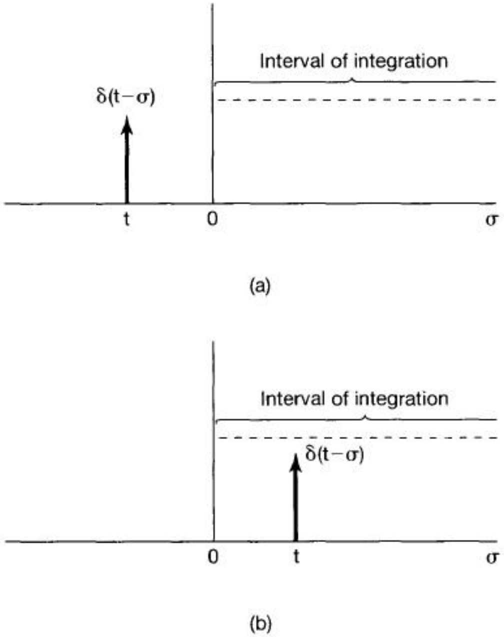


Figure 1.38 Relationship given in eq. (1.75): (a) $t < 0$; (b) $t > 0$.

1) Memoryless or With Memory

Memoryless : output at a given time depends only on the input at the same time

eg. $$y[n] = (ax[n] - x^2[n])^2$$

With Memory

eg. $$y[n] = \sum_{k=-\infty}^{n} x[k]$$

summer or accumulator

Memoryless or With Memory (cont.)

- In physical system, memory is associated with the storage of energy, e.g., capacitor in electric circuit.

2) Invertability

invertible : distinct inputs lead to distinct outputs, i.e. an inverse system exits

No information loss

eg.

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$

$$z[n] = y[n] - y[n-1] \ = x[n]$$

3) Causality

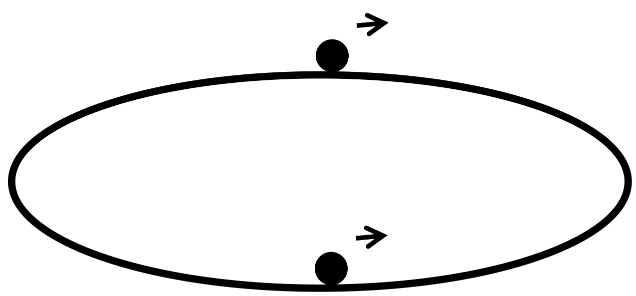
- **Causality**: A system is causal if the output does **not** anticipate future values of the input, i.e., if the output at any time depends only on values of the input up to that time.
- All real-time physical systems are **causal**, because time only moves forward, and effect occurs after cause. (Imagine if you own a noncausal system whose output depends on tomorrow's stock price.)
  - Do **not** apply to spatially varying signals. (We can move both left and right, up and down.)
  - Do **not** apply to systems processing *recorded (or non-realtime)* signals, e.g. taped sports games vs. live broadcast.

Causal or Non-causal?

- $y(t) = x^2(t-1)$
- $y(t)=x(t+1)$
- $y(t)=x(t) \cos(t+1)$
- $y[n]=x[-n]$
- $y[n]=(1/2)^{n+1} x^3[n-1]$

4) Stability

- Small input leads to response that does not diverge.



- If the input to a stable system is bounded, the output must also be bounded.
- e.g.: S_1: $y(t) = t\,x(t)$

  S_2: $y(t) = e^{x(t)}$

Stability (cont.)

- Stability of physical systems are due to the presence of mechanisms that dissipate energy, e.g., resistor, friction.
- Stable or not (Example 1.13 in textbook)?
  - Find specific example, e.g, for S: $y(t)=t\,x(t)$
  - To prove that all bounded inputs lead to bounded output.

5) Time Invariance (TI)

- Informally, a system is time-invariant (TI) if its behavior does not depend on the choice of $t = 0$. Then two identical experiments will yield the same results, regardless the starting time.
  - Mathematically (in DT): A system $x[n] \to y[n]$ is TI if for *any* input $x[n]$ and *any* time shift n_0

$$\begin{array}{ll} \text{If} & x[n] \to y[n] \\ \text{then} & x[n-n_0] \to y[n-n_0]\,. \end{array}$$

  - Similarly for CT time-invariant system

$$\begin{array}{ll} \text{If} & x(t) \to y(t) \\ \text{then} & x(t-t_0) \to y(t-t_0)\,. \end{array}$$

Time-invariant or Time-varying?

- **Steps:**

1) **Calculate $y_1(t) \leftarrow x_1(t)$**
2) **Calculate $y_2(t) \leftarrow x_2(t)=x_1(t-t_0)$**
3) **Does $y_1(t-t_0)$ equal $y_2(t)$?**

e.g.: $y[n] = \left(\frac{1}{2}\right)^{n+1} x^3[n-1]$

① $y_1[n] = \left(\frac{1}{2}\right)^{n+1} x_1^3[n-1]$

② $x_2[n] = x_1[n-n_0]$

$y_2[n] = \left(\frac{1}{2}\right)^{n+1} x_2^3[n-1]$

$= \left(\frac{1}{2}\right)^{n+1} x_1^3[n-n_0-1]$

③ $y_1[n-n_0] = \left(\frac{1}{2}\right)^{n-n_0+1} \cdot x_1^3[n-n_0-1]$

$\therefore$ $y_1[n-n_0] \neq y_2[n]$

$\therefore$ Time-varying

Now we can deduce something:

- If the input to a TI system is periodic, then the output is also periodic with the same period (Problem 1.43 (a)).

Proof: Suppose $x(t+T) = x(t)$

and $x(t) \rightarrow y(t)$

Then by TI

$$x(t+T) \rightarrow y(t+T)$$

$$\uparrow \qquad \qquad \uparrow$$

But these are the same input!

So these must be the same output, *i.e.*, $y(t) = y(t+T)$

6) Linear and Nonlinear Systems

- Many systems are nonlinear.
  - **e.g.:** economic system with
    input: fiscal and monetary policies, labor, resources, etc.
    → output: GDP, inflation, etc.
  - System behaviour is very unpredictable, because it is highly nonlinear.
- We will deal with only linear systems, which are good approximations of nonlinear systems in certain ranges.
  - **e.g.:** small-signal conductance of a nonlinear diode.
- Linear systems can be analyzed accurately.

Linearity

We have $x_1(t) \rightarrow y_1(t)$, and $x_2(t) \rightarrow y_2(t)$.

The system is linear, if:

1) **Additivity property:** $x_1(t) + x_2(t) \rightarrow y_1(t) + y_2(t)$
2) **Scaling (or homogeneity) property:**

$$a\, x_1(t) \rightarrow a\, y_1(t)$$

where a is a complex number

e.g.: $y(t) = 2\, x(t)$ $\quad\quad$ $y(t) = x^2(t)$

Linearity (cont.)

A (CT) system is linear if it has the superposition property:

If $\quad x_1(t) \to y_1(t) \quad$ and $\quad x_2(t) \to y_2(t)$

then $\quad ax_1(t) + bx_2(t) \;\to\; ay_1(t) + by_2(t)$

- For linear systems, zero input $\to$ zero output

"Proof" $\quad 0 = 0 \cdot x[n] \to 0 \cdot y[n] = 0$

Question: Is the system $y = A + x$ linear?

- *Incrementally linear system!*

Linear system or not?

- **Steps**

1) Have $y_1(t)$ and $y_2(t)$ as output signals to $x_1(t)$ and $x_2(t)$

2) Have $y_3(t)$ as output signal to $x_3(t) = a\, x_1(t) + b\, x_2(t)$

3) Does $y_3(t)$ equal "$a\, y_1(t) + b\, y_2(t)$"?

More examples on textbook

Read Example 1.17 ~ 1.20

Linearity (cont.)

- Superposition

$$\text{If} \qquad x_k[n] \rightarrow y_k[n]$$

$$\text{Then} \qquad \boxed{\sum_k a_k x_k[n] \rightarrow \sum_k a_k y_k[n]}$$

- This property seems to be almost trivial now, but it is one of the most important ones

Linear Time-Invariant (LTI) Systems

- Focus of most of this course
  - Practical importance
  - The powerful analysis tools (*transformation*) associated with LTI systems
- A basic fact: If we know the response of an LTI system to some inputs, we actually know the response to many inputs.

Exploiting Superposition and Time-Invariance

If we have $x_k[n] \rightarrow y_k[n]$, then

$$x[n] = \sum_k a_k x_k[n] \xrightarrow{\text{Linear System}} y[n] = \sum_k a_k y_k[n]$$

Question: Are there sets of "basic" signals $x_k[n]$ such that

a) We can represent *any* signals as linear combinations of these building block signals.

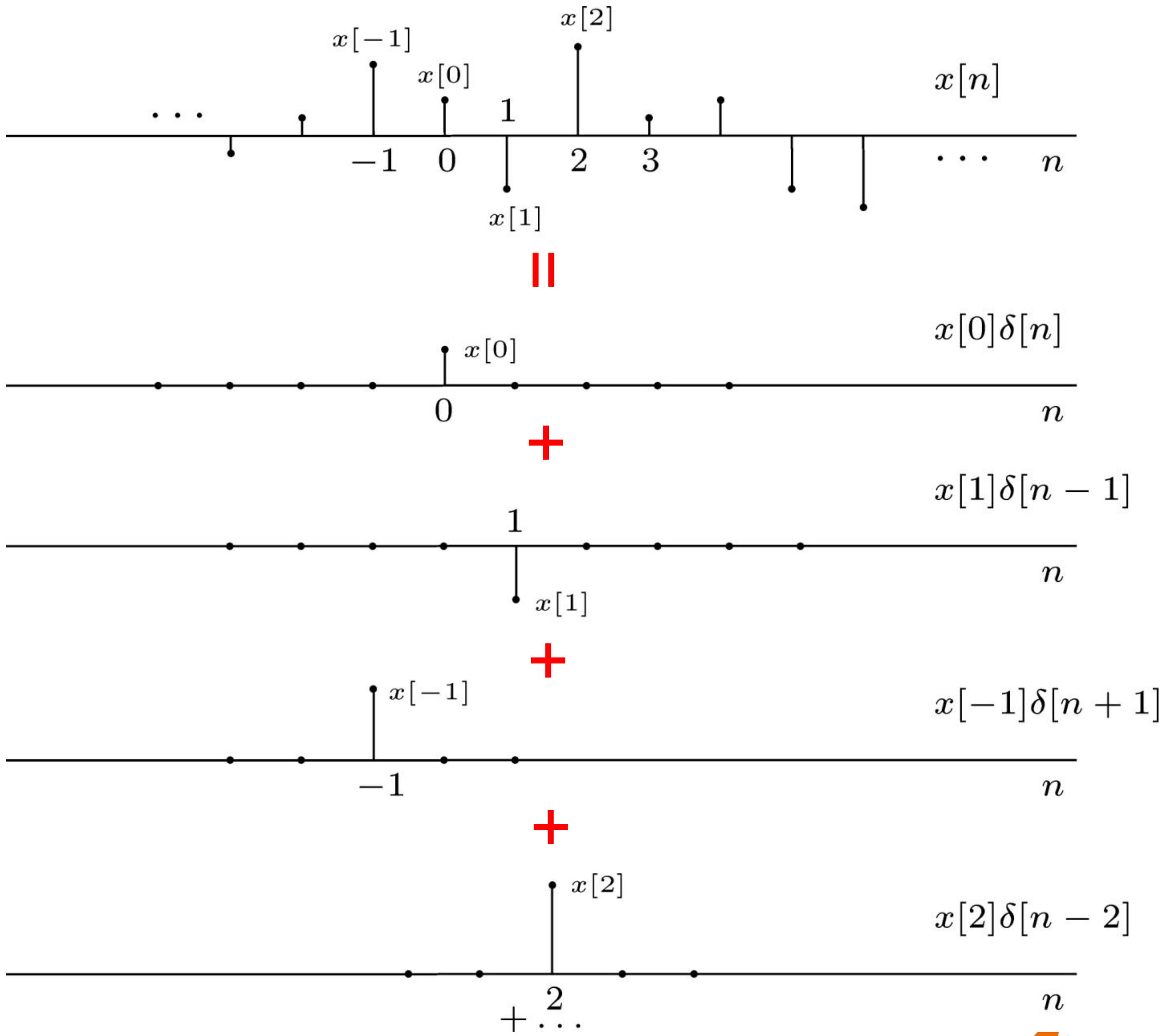
b) The response of LTI Systems to these basic signals are both *simple* and *insightful*.

Fact: For LTI Systems (CT or DT) there are two natural choices for these building blocks

Focus for now: DT Shifted unit samples $\delta[n-n_o]$

Next time: CT Shifted unit impulses $\delta(t-t_o)$

Representation of DT Signals Using Unit Samples



That is ...

$$x[n] = \ldots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \ldots$$

$$\Downarrow$$

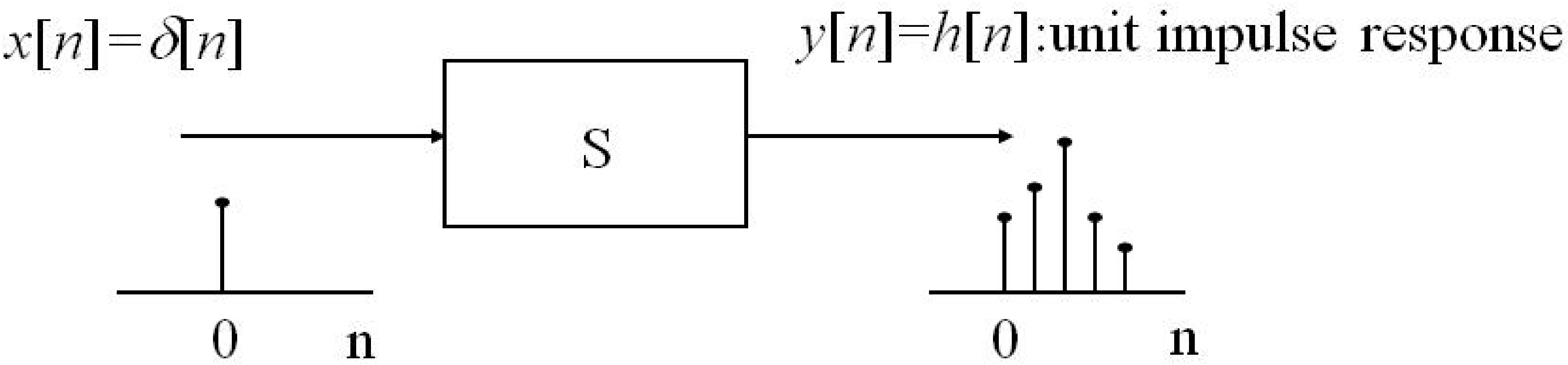
Important to note the "-" sign

$$x[n] = \sum_{k=-\infty}^{+\infty} \underbrace{x[k]}_{\text{Coefficients}}\underbrace{\delta[n-k]}_{\text{Basic Signals}}$$

– **Sifting property** of the unit impulse: looked at the index k, $\delta[n\text{-}k]$ is nonzero only at $k = n$, which "sifts" the value $x[n]$ out of the function $x[k]$.

Unit Impulse Response (Unit Sample Response)

- Define the output for an unit impulse input as the unit impulse response



Example: $y[n]=x[n]+2x[n\text{-}1]+4x[n\text{-}2]$

What is unit impulse response?

Response of DT LTI Systems

$$x[n] \longrightarrow \boxed{\text{DT System}} \longrightarrow y[n]$$

- Now suppose the system is **LTI**, and define the *unit impulse response* $h[n]$:

$$\delta[n] \longrightarrow h[n]$$

$$\Downarrow$$

From Time-Invariance:

$$\delta[n-k] \longrightarrow h[n-k]$$

From Linearity:

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k]\,\delta[n-k] \longrightarrow y[n] = \underbrace{\sum_{k=-\infty}^{+\infty} x[k]\,h[n-k]}_{\textit{convolution sum}} = x[n] * h[n]$$

The output for an arbitrary input signal is the superposition of a series of "shifted, scaled unit impulse response"

$$y[n]=\sum_{k=-N}^{N} x[k]h[n-k]$$

$$\mathbf{y}=\mathbf{H}\mathbf{x}$$

$$\begin{bmatrix} y[-N] \\ y[-N+1] \\ \vdots \\ y[0] \\ \vdots \\ y[N-1] \\ y[N] \end{bmatrix} = \begin{bmatrix} h[0] & h[-1] & \cdots & h[-N] & \cdots & h[-2N+1] & h[-2N] \\ h[1] & h[0] & \cdots & h[-N+1] & \cdots & h[-2N+2] & h[-2N+1] \\ \vdots & \vdots & \cdots & \vdots & \cdots & \vdots & \vdots \\ h[N] & h[N-1] & \cdots & h[0] & \cdots & h[-N+1] & h[-N] \\ \vdots & \vdots & \cdots & \vdots & \cdots & \vdots & \vdots \\ h[2N-1] & h[2N-2] & \cdots & h[N-1] & \cdots & h[0] & h[-1] \\ h[2N] & h[2N-1] & \cdots & h[N] & \cdots & h[1] & h[0] \end{bmatrix} \begin{bmatrix} x[-N] \\ x[-N+1] \\ \vdots \\ x[0] \\ \vdots \\ x[N-1] \\ x[N] \end{bmatrix}$$

Toeplitz matrix

$$H_{i,j}=H_{i+1,j+1}=h_{i-j}$$

Convolution operation procedure:

$$h[k] \xrightarrow{\text{Flip}} h[-k] \xrightarrow{\text{Slide}} h[n-k] \xrightarrow{\text{Multiply}} x[k]h[n-k]$$

$$\xrightarrow{\text{Sum}} \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

F S M S

$$\Downarrow$$

$$y[n]$$

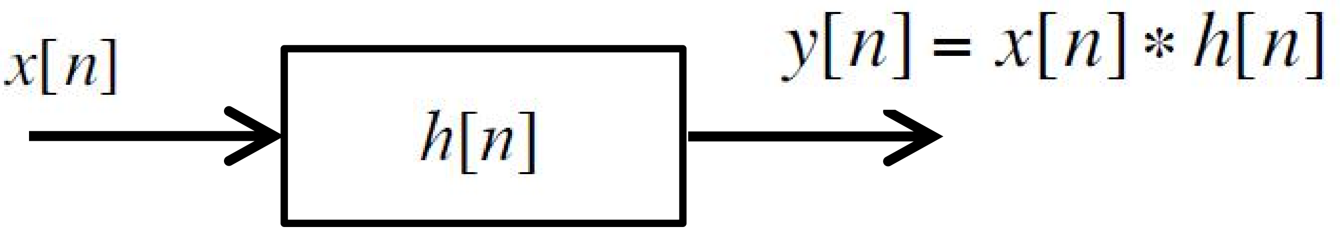
$$\text{Any DT LTI} \longleftrightarrow y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{+\infty} x[k]\, h[n-k]$$

$x[n]$: input

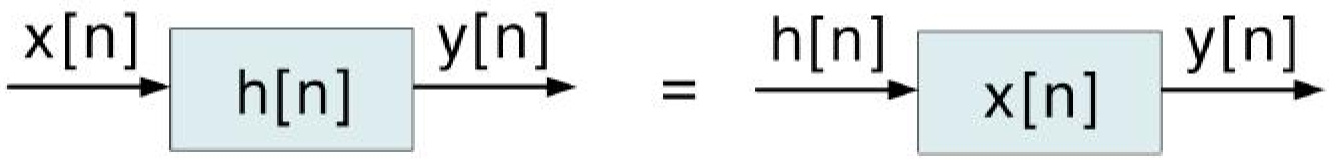
$y[n]$: output

$h[n]$: impulse response of the system



The Commutative Property of Convolution

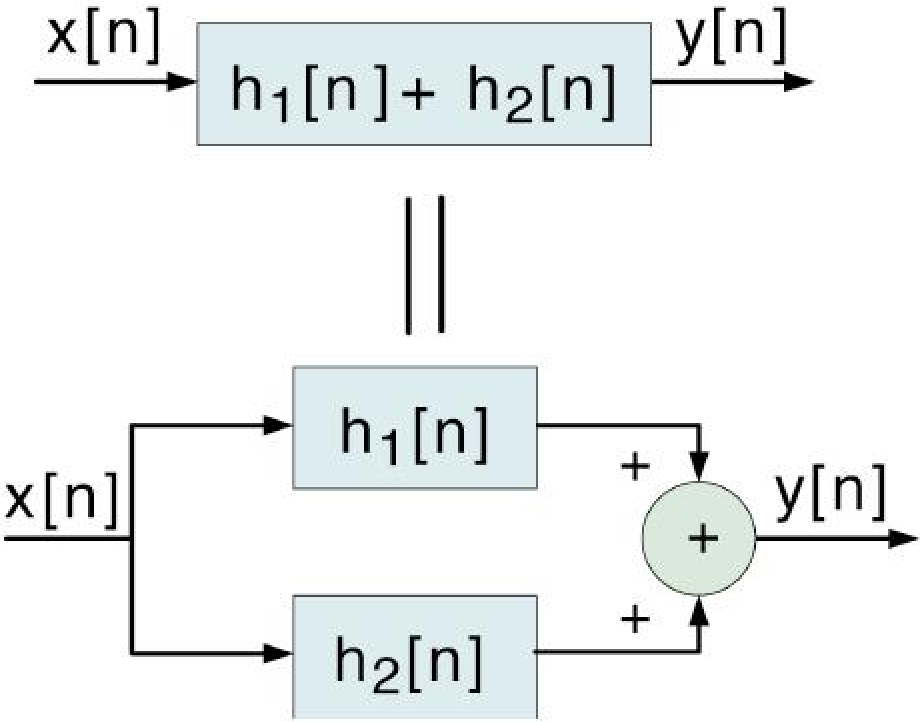
$$y[n] = x[n] * h[n] \quad = \quad h[n] * x[n]$$



The Distributive Property of Convolution

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

Interpretation



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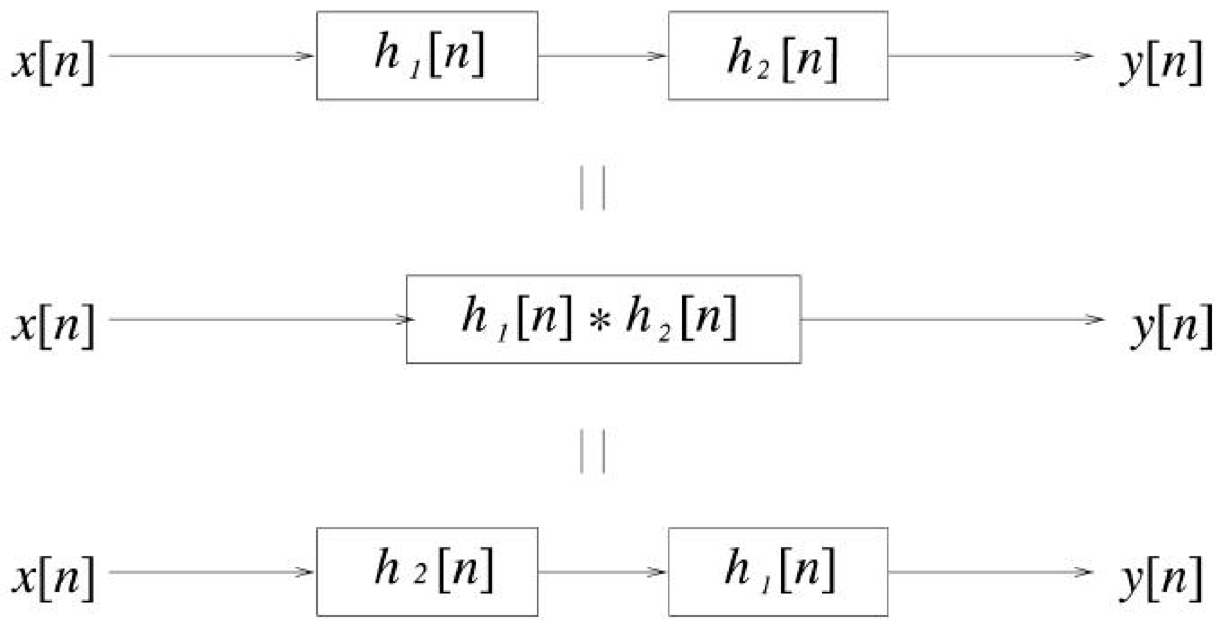
The Associative Property of Convolution

$$x[n]*(h_1[n]*h_2[n]) = (x[n]*h_1[n])*h_2[n]$$

(Commutativity) $\qquad \|$

$$x[n]*(h_2[n]*h_1[n]) = (x[n]*h_2[n])*h_1[n]$$

Implication (Very special to LTI Systems)



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Properties of Convolution

Combining the Commutative property,

$$y[n] = x[n] * h[n] = h[n] * x[n]$$

Distributive property,

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

and Associative property,

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

symbolically, we can treat "$*$" as a "$\times$". Easy, piece of cake!

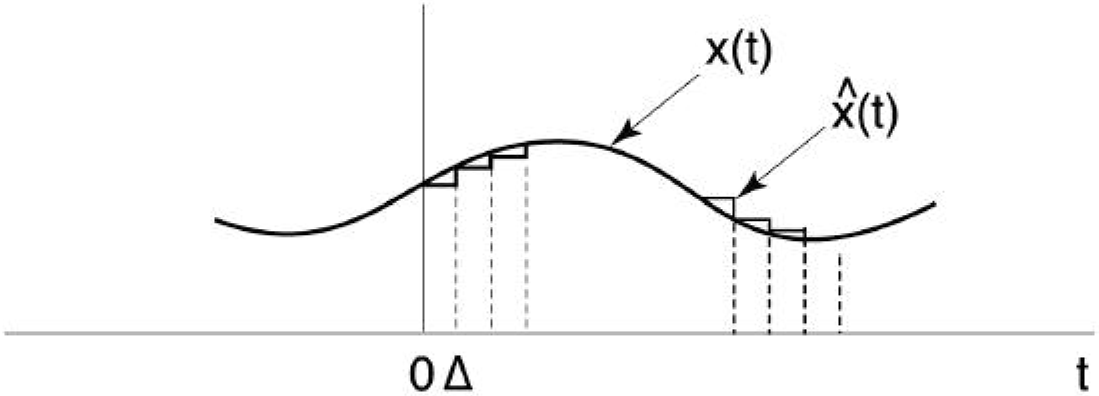
The hard part is the actual calculation of the convolution.

Flip → Slide → Multiply → Sum.

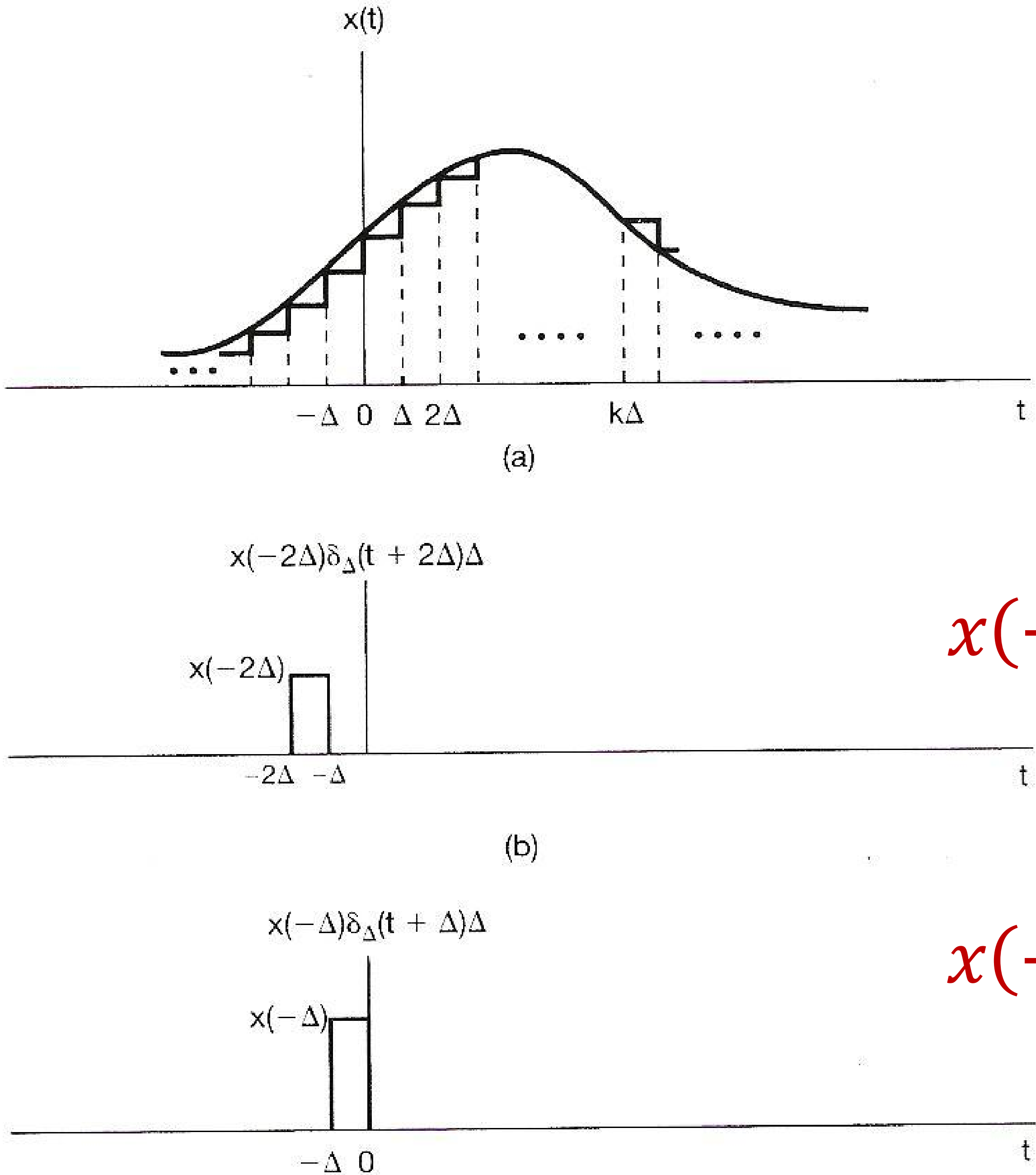
Soon we will develop a clever way (*transformation*) to perform "$\times$" instead of "$*$" operation.

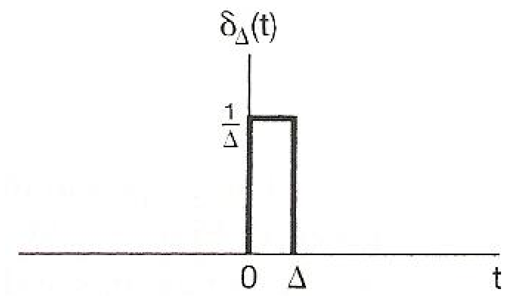
Representation of CT Signals

- Approximate any input $x(t)$ as a sum of shifted, scaled pulses (in fact, that is how we do integration)



$$x(t) = \lim_{\Delta \to 0} \hat{x}(t)$$

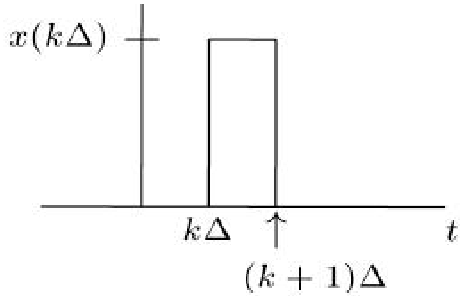




$$x(-2\Delta)\delta_\Delta(t+2\Delta)\Delta$$

$$x(-\Delta)\delta_\Delta(t+\Delta)\Delta$$

Representation of CT Signals (cont.)



$$= \quad x(k\Delta)\delta_\Delta(t - k\Delta)\Delta$$

$$\Downarrow$$

$$\hat{x}(t) = \sum_{k=-\infty}^{+\infty} x(k\Delta)\delta_\Delta(t - k\Delta)\Delta$$

$$\downarrow \quad \text{limit as } \Delta \to 0$$

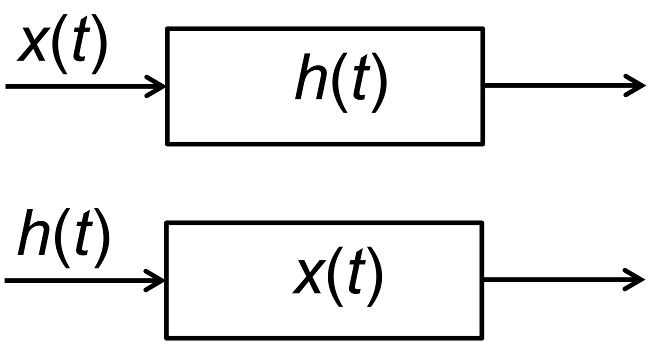
$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau}$$

Sifting property of the unit impulse

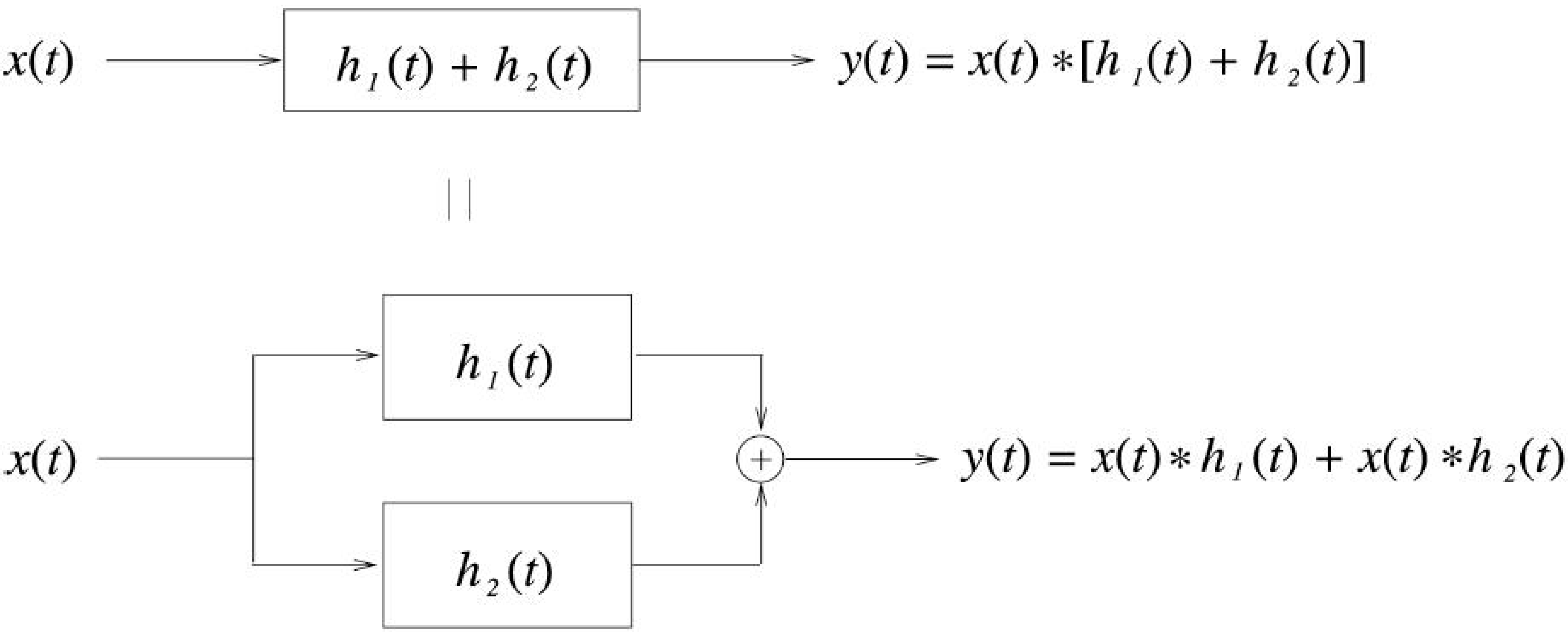
Property: Commutative

$$x(t) * h(t) = h(t) * x(t)$$

- The role of input signal and unit impulse response is interchangeable, giving the same output signal



Property: Distributive



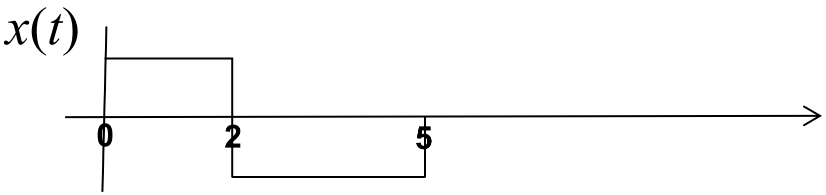
Property: Distributive (Cont.)

$$[x_1(t)+x_2(t)]*h(t)=x_1(t)*h(t)+x_2(t)*h(t)$$

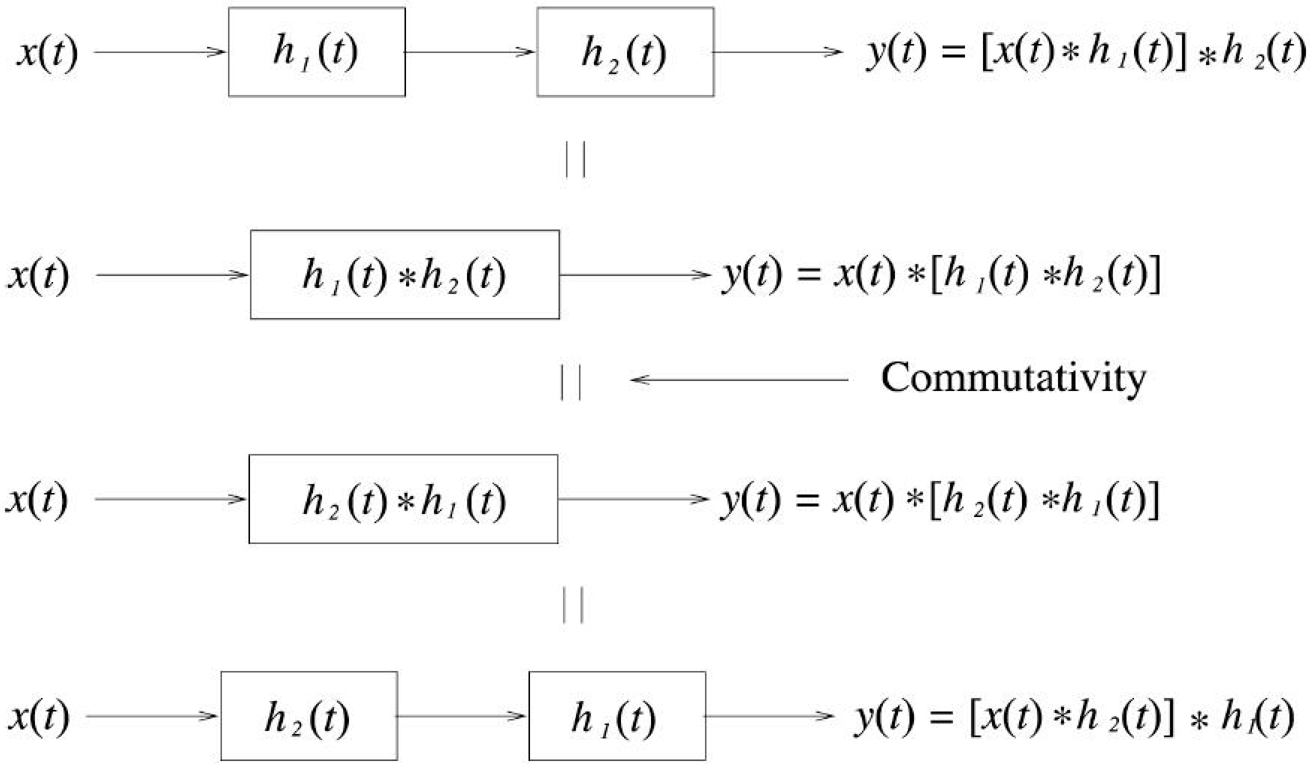
Problem 2.22 (b)

$$x(t)=u(t)-2u(t-2)+u(t-5)$$

$$h(t)=e^{2t}u(1-t)$$

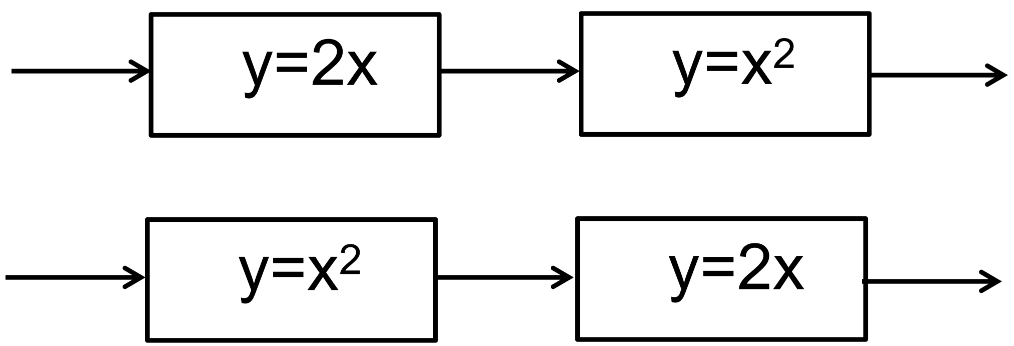


Properties: Associative



Properties: Associative (Cont.)

- The order in which non-linear systems are cascaded cannot be changed.
- e.g.



Property: Memory/Memoryless

– A linear, time-invariant, causal system is memoryless only

if $h[n] = K\delta[n]$ $\quad h(t) = K\delta(t)$

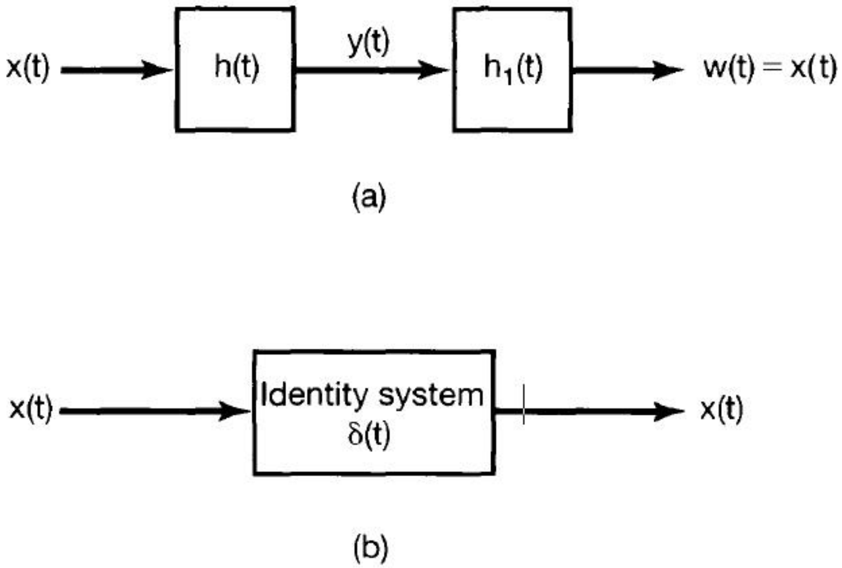
$y[n] = Kx[n]$ $\quad y(t) = Kx(t)$

if k=1 further, they are identity systems

$$y[n] = x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n] * \delta[n]$$

$$y(t) = x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = x(t) * \delta(t)$$

Property: Invertibility



Property: Causality

Causality: CT LTI system is causal $\leftrightarrow$ $h(t) = 0$, at $t < 0$

- This is because that the input unit impulse function δ(t)=0 at t<0

$t - \tau \geq 0$, or $\tau \leq t$

As a result:

$$y(t) = \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau = \int_{-\infty}^{t} x(\tau)h(t-\tau)d\tau$$

$y(t)$ only depends on $x(\tau < t)$.

Property: Stability

BIBO Stability: CT LTI system is stable $\Leftrightarrow \int_{-\infty}^{+\infty} |h(\tau)|\, d\tau < \infty$

→ Sufficient condition:

For $|x(t)| \le x_{\max} < \infty$.

$$|y(t)| = \left|\int_{-\infty}^{+\infty} x(\tau)h(t-\tau)d\tau\right| \le x_{\max}\left|\int_{-\infty}^{+\infty} h(t-\tau)d\tau\right| < \infty.$$

→ Necessary condition:

Suppose $\int_{-\infty}^{+\infty} |h(\tau)|d\tau = \infty$

Let $x(t) = h^*(-t)/|h^*(-t)|$, then $|x(t)| \equiv 1$ bounded

$$\text{But } y(0) = \int_{-\infty}^{+\infty} x(\tau)h(-\tau)d\tau = \int_{-\infty}^{+\infty} \frac{h^*(-\tau)h(-\tau)}{|h(-\tau)|}d\tau = \int_{-\infty}^{+\infty} |h(-\tau)|d\tau = \infty$$

Unit Step Response

unit step function → unit step response

Step response

$$s(t) = u(t) * h(t) = h(t) * u(t) = \int_{-\infty}^{t} h(\tau)\,d\tau$$

$$x(t)=\delta(t) \longrightarrow \boxed{h(t)} \longrightarrow$$ $h(t)$: unit impulse response

$$x(t)=u(t) \longrightarrow \boxed{h(t)} \longrightarrow$$ $s(t)$: unit step response

The relation between unit step function and unit impulse function

$$h(t) = \frac{ds(t)}{d(t)} = s'(t)$$

Integrators

$$x(t) \longrightarrow \boxed{\int} \longrightarrow y(t) = \int_{-\infty}^{t} x(\tau)d\tau$$

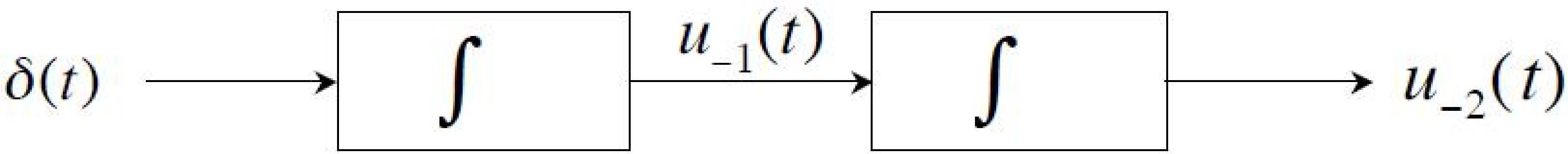
Impulse response: $\boxed{u_{-1}(t) \equiv u(t)}$

Operational definition: $x(t) * u_{-1}(t) = \int_{-\infty}^{t} x(\tau)d\tau$

Cascade of n integrators:

$$u_{-n}(t) = \underbrace{u_{-1}(t) * \cdots * u_{-1}(t)}_{n\ times} \qquad (n > 0)$$

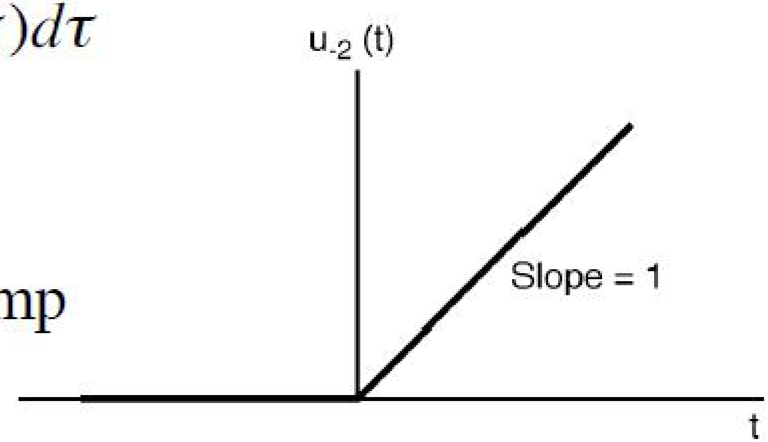
Integrators (Cont.)



$$u_{-2}(t) = \int_{-\infty}^{t} u_{-1}(\tau)d\tau = \int_{-\infty}^{t} u(\tau)d\tau$$

$$= u(t)\int_{0}^{t} d\tau$$

$$= t \cdot u(t) \qquad \text{the unit ramp}$$



More generally, for $n > 0$

$$u_{-n}(t) = \frac{t^{(n-1)}}{(n-1)!}u(t)$$

Notation

Define $\boxed{u_0(t) = \delta(t)}$

Then $$u_n(t) * u_m(t) = u_{n+m}(t)$$

n and m can be $\pm$

E.g. $$u_1(t) * u_{-1}(t) = u_0(t)$$

$$||$$

$$\left(\frac{d}{dt}u(t)\right) = \delta(t)$$

Overview on Frequency Analysis

	Fourier Series (Periodic/Discrete)	**Fourier Transform (Aperiodic/Continuous)**
Continuous-Time Domain	CT Fourier Series	CT Fourier Transform
Discrete-Time Domain	DT Fourier Series	DT Fourier Transform

Eigenfunctions $\phi_k(t)$ and Their Properties

- (Focus on CT systems now, but results apply to DT systems as well.)

$$\phi_k(t) \longrightarrow \boxed{\text{System}} \longrightarrow \underbrace{\lambda_k}_{\text{eigenvalue}} \ \underbrace{\phi_k(t)}_{\text{eigenfunction}}$$

Eigenfunction in → same function out with a “gain”

- From the superposition property of LTI systems:

$$x(t) = \sum_k a_k \phi_k(t) \longrightarrow \boxed{\text{LTI}} \longrightarrow y(t) = \sum_k \lambda_k a_k \phi_k(t)$$

Now the task of finding response of LTI systems is to determine λ_k.
The solution is simple, general, and insightful.

Two Key Questions

1. What are the eigenfunctions of a general LTI system?

2. What kinds of signals can be expressed as superpositions of these eigenfunctions?

Complex Exponentials - The *Only* Eigenfunctions of *Any* LTI Systems

$$x(t) = e^{st} \longrightarrow \boxed{h(t)} \longrightarrow y(t) = \int_{-\infty}^{+\infty} h(\tau) e^{s(t-\tau)} d\tau$$

$$= \left[\int_{-\infty}^{+\infty} h(\tau) e^{-s\tau} d\tau\right] e^{st}$$

$$= \underbrace{H(s)}_{\text{eigenvalue}} \underbrace{e^{st}}_{\text{eigenfunction}}$$

$$\boxed{y(t) = x(t) * h(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t-\tau) d\tau}$$

$$x[n] = z^n \longrightarrow \boxed{h[n]} \longrightarrow y[n] = \sum_{m=-\infty}^{\infty} h[m] z^{n-m}$$

$$= \left[\sum_{m=-\infty}^{\infty} h[m] z^{-m}\right] z^n$$

$$= \underbrace{H(z)}_{\text{eigenvalue}} \underbrace{z^n}_{\text{eigenfunction}}$$

System Functions $H(s)$ or $H(z)$

CT: $$x(t) \xrightarrow{e^{st}} \boxed{h(t)} \xrightarrow{H(s)e^{st}} y(t)$$

$$H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}dt$$

$$x(t) = \sum a_k e^{s_k t} \longrightarrow y(t) = \sum H(s_k)a_k e^{s_k t}$$

DT: $$x[n] \xrightarrow{z^n} \boxed{h[n]} \xrightarrow{H(z)z^n} y[n]$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n]z^{-n}$$

$$x[n] = \sum a_k z_k^n \longrightarrow y[n] = \sum H(z_k)a_k z_k^n$$

CT Fourier Series Pairs

$$(\omega_o = \frac{2\pi}{T})$$

Review:

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{+\infty} a_k e^{j2\pi kt/T}$$

$$a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\,dt$$

Skip it in future for shorthand

$$x(t) \xleftrightarrow{FS} a_k$$

(A Few) Properties of CT Fourier Series

- Linearity $x(t) \longleftrightarrow a_k,\ y(t) \longleftrightarrow b_k, \Rightarrow \alpha x(t) + \beta y(t) \longleftrightarrow \alpha a_k + \beta b_k$
- Conjugate Symmetry

$$x(t) \quad real \quad \Rightarrow \quad a_{-k} = a_k^{\,*}$$

Proof: $a_{-k} = \frac{1}{T}\int_T x(t)e^{jk\omega_o t}dt = \left[\frac{1}{T}\int_T x^*(t)e^{-jk\omega_o t}dt\right]^* = a_k^*$

$$\Downarrow\ a_k = \mathrm{Re}\{a_k\} + j\,\mathrm{Im}\{a_k\}$$

$$\boxed{\mathrm{Re}\{a_{-k}\} + j\,\mathrm{Im}\{a_{-k}\}} \qquad = |a_k|e^{j\angle a_k} \qquad \boxed{\mathrm{Re}\{a_k\} - j\,\mathrm{Im}\{a_k\}}$$

$\mathrm{Re}\{a_k\}$ is even , $\mathrm{Im}\{a_k\}$ is odd

or $|a_k|$ is even, $\angle a_k$ is odd

- Time shift

$$x(t) \longleftrightarrow a_k$$

$$x(t-t_o) \longleftrightarrow a_k e^{-jk\omega_o t_o} = a_k e^{-jk2\pi t_o/T}$$

Introduce a linear phase shift $\propto t_o$

- Time Reversal

$$x(-t) \xleftrightarrow{FS} a_{-k}$$

the effect of sign change for $x(t)$ and a_k are identical

Example: $x(t)$: $\ldots\ a_{-2}\ a_{-1}\ a_0\ a_1\ a_2\ \ldots$

$x(-t)$: $\ldots\ a_2\ a_1\ a_0\ a_{-1}\ a_{-2}\ \ldots$

- Time Scaling

α : positive real number

$x(\alpha t)$: periodic with period T/α and fundamental frequency αw_0

$$x(\alpha t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(\alpha w_0)t}$$

a_k unchanged, but $x(\alpha t)$ and each harmonic component are different

$$\begin{aligned} c_k &= \frac{1}{T}\int_T \sum_n \sum_l a_n b_l e^{j(n+l)\omega_0 t} e^{-jk\omega_0 t} dt \\ &= \frac{1}{T}\sum_n \sum_l a_n b_l \delta(k-(n+l)) \\ &= \sum_n a_n b_{k-n} \end{aligned}$$

- **Multiplication Property**

$x(t) \longleftrightarrow a_k$, $y(t) \longleftrightarrow b_k$ (Both $x(t)$ and $y(t)$ are periodic with the same period T)

$$\Downarrow$$

$$x(t)\cdot y(t) \longleftrightarrow c_k = \sum_{l=-\infty}^{+\infty} a_l b_{k-l} = a_k * b_k$$

Proof: $$\underbrace{\sum_l a_l e^{jl\omega_o t}}_{x(t)} \cdot \underbrace{\sum_m b_m e^{jm\omega_o t}}_{y(t)} = \sum_{l,m} a_l b_m e^{j(l+m)\omega_o t} \xrightarrow{l+m=k} \sum_k \underbrace{\left[\sum_l a_l b_{k-l}\right]}_{c_k} e^{jk\omega_o t}$$

- Parseval Relation

$$\underbrace{\frac{1}{T}\int_T |x(t)|^2 dt}_{\text{Average Signal Power}} = \sum_{k=-\infty}^{\infty} \underbrace{|a_k|^2}_{\text{Power of component } e^{jk\omega_0 t}}$$

Observation: power is the same whether measured in the time-domain or the frequency-domain

Proof:

$$\frac{1}{T}\int_T |x(t)|^2 dt = < \sum_{l=-\infty}^{+\infty} a_l e^{jlw_0 t}, \sum_{k=-\infty}^{+\infty} a_k e^{jkw_0 t} >$$

$$= \sum_{k=-\infty}^{+\infty} < a_k e^{jkw_0 t}, a_k e^{jkw_0 t} > = \sum_{k=-\infty}^{+\infty} |a_k|^2$$

More

Note:
$x(t)$ is periodic with fundamental frequency ω_0

$$x(t) \longleftrightarrow a_k$$

- **Frequency shifting**

$$e^{jM\omega_0 t}x(t) \longleftrightarrow a_{k-M}$$

- **Differentiation**

$$\frac{dx}{dt} \longleftrightarrow jk\omega_0 a_k$$

- **Integration**

$$\int_{-\infty}^{t} x(t)dt \longleftrightarrow (\frac{1}{jk\omega_0})a_k$$

DT Fourier Series Pair $\left(\omega_o = \frac{2\pi}{N}\right)$

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} \qquad \text{(Synthesis equation)}$$

$$a_k = \frac{1}{N} \sum_{n=\langle N \rangle} x[n] e^{-jk\omega_0 n} \qquad \text{(Analysis equation)}$$

Different from CT Fourier series

DT Fourier Series - Properties

- Strong similarities between the properties of DT and CT Fourier series [Comparing Table 3.2 to Table 3.1.]
- For example
  - Fourier series of $x[n-n_0]$, $e^{jm\omega_0 n}x[n]$
  - Fourier series properties for real signal
  - Fourier series of the multiplication

Two Important Properties

- **Periodic convolution:**

  Suppose x and y are two periodic signals with common period N, the periodic convolution between x and y is defined as

$$x[n] \circledast y[n] = \sum_{k=<N>} x[k]y[n-k]$$

- **Suppose $x[n] \rightarrow a_k$ and $y[n] \rightarrow b_k$, then**

$$x[n] \circledast y[n] \rightarrow \mathrm{N}\mathrm{a}_\mathrm{k}\mathrm{b}_\mathrm{k} \quad \text{and} \quad x[n]y[n] \rightarrow a_k \circledast b_k$$

- **Parseval's Relation**

$$\frac{1}{N} \sum_{n=<N>} |x[n]|^2 = \sum_{k=<N>} |a_k|^2$$

The CT Fourier Transform Pair

$$X(j\omega)=\int_{-\infty}^{+\infty}x(t)e^{-j\omega t}dt \quad \text{— } FT$$

Fourier Transform

$$x(t)=\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\omega)e^{j\omega t}d\omega \quad \text{— Inverse } FT$$

Inverse Fourier Transform

$$x(t)\overset{\mathbf{F}}{\longleftrightarrow}X(j\omega)$$

$$\mathbf{F}\,(x(t))=X(j\omega)$$

$$x(t)=\mathbf{F}^{-1}(X(j\omega))$$

CT Fourier Transforms of Periodic Signals

Suppose $$X(j\omega) = \delta(\omega - \omega_0)$$

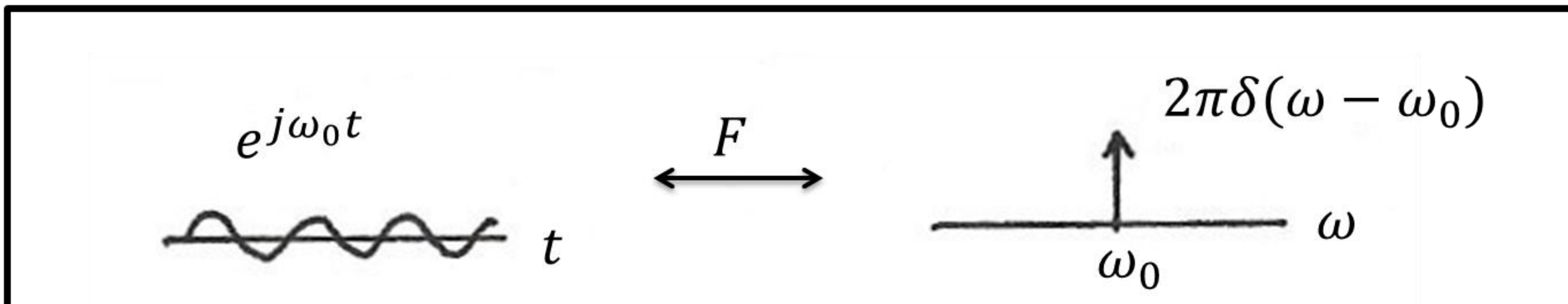
$$\Downarrow$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega = \frac{1}{2\pi} e^{j\omega_0 t}$$

— periodic in t with frequency ω_o

That is $$\boxed{e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)}$$

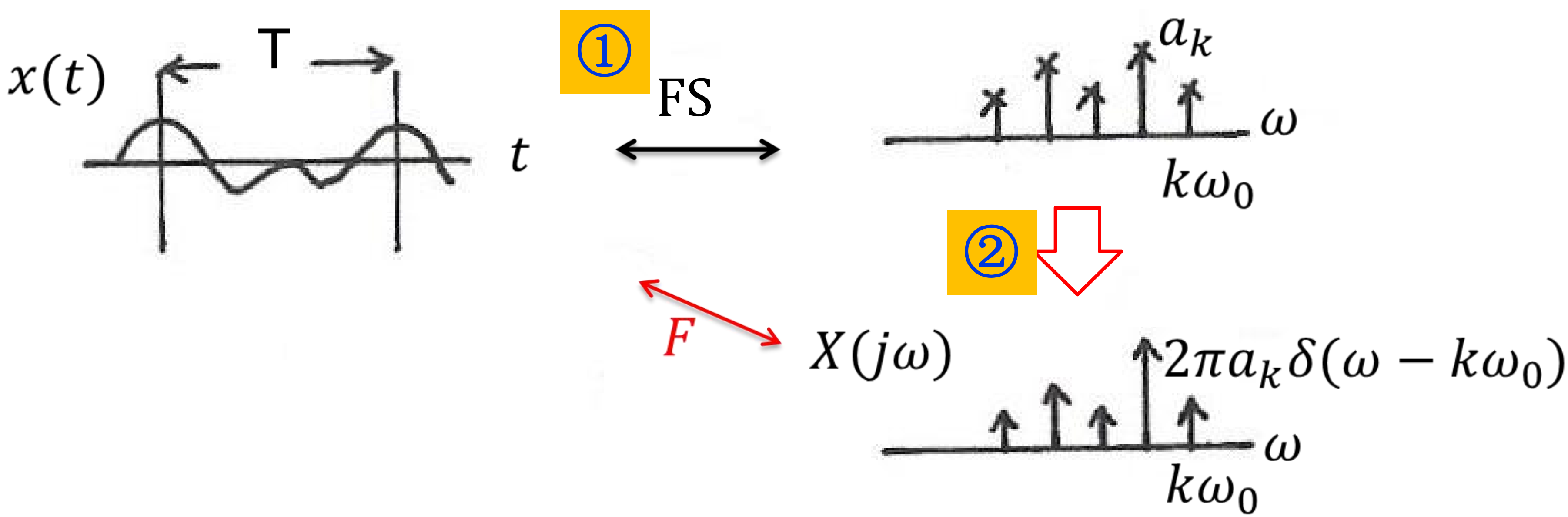
— All the energy is concentrated in one frequency — ω_o



Fourier Transform for Periodic Signals – Unified Framework

More generally, if $x(t) = x(t+T)$, then

$$x(t) = \sum_{k=-\infty}^{+\infty} a_k e^{jk\omega_0 t} \longleftrightarrow X(j\omega) = \sum_{k=-\infty}^{+\infty} 2\pi a_k \delta(\omega - k\omega_0)$$ Discrete spectra



Properties of the CT Fourier Transform

1) Linearity

$$x(t) \xleftrightarrow{F} X(j\omega),\ y(t) \xleftrightarrow{F} Y(j\omega)$$

$$ax(t) + by(t) \xleftrightarrow{F} aX(j\omega) + bY(j\omega)$$

2) Time Shifting

$$x(t - t_0) \longleftrightarrow e^{-j\omega t_o} X(j\omega)$$

Proof: $\int_{-\infty}^{\infty} x(\underbrace{t - t_o}_{t'})e^{-j\omega t}dt = e^{-j\omega t_o}\underbrace{\int_{-\infty}^{\infty} x(t')e^{-j\omega t'}dt'}_{X(j\omega)}$

FT magnitude unchanged

$$\left|e^{-j\omega_0 t}X(j\omega)\right| = \left|X(j\omega)\right|$$

Linear change in *FT* phase

$$\angle(e^{-j\omega_0 t}X(j\omega)) = \angle X(j\omega) - \omega t_0$$

CTFT Properties (cont.)

3) Conjugation & Conjugate Symmetry

- Conjugation $x^*(t) \xleftrightarrow{F} X^*(-j\omega)$

- Conjugate Symmetry $x(t)$ real $\longleftrightarrow X(-j\omega) = X^*(j\omega)$

$|X(-j\omega)| = |X(j\omega)|$ *Even* Or $\text{Re}\{X(-j\omega)\} = \text{Re}\{X(j\omega)\}$ *Even*

$\angle X(-j\omega) = -\angle X(j\omega)$ *Odd* $\text{Im}\{X(-j\omega)\} = -\text{Im}\{X(j\omega)\}$ *Odd*

When $x(t)$ is real (all the physically measurable signals are *real*), the negative frequency components do *not* carry any additional information from the positive frequency components. $\omega \geq 0$ will be sufficient.

CT Fourier Series Property

- Conjugate Symmetry

$$x(t) \quad real \quad \Rightarrow \quad a_{-k} = a_k{}^*$$

Proof: $$a_{-k} = \frac{1}{T}\int_T x(t)e^{jk\omega_0 t}dt = \left[\frac{1}{T}\int_T x^*(t)e^{-jk\omega_0 t}dt\right]^* = a_k^*$$

$\because$

$$a_k = \mathrm{Re}\{a_k\} + j\mathrm{Im}\{a_k\}$$

$\therefore$

$$\boxed{\mathrm{Re}\{a_{-k}\} + j\,\mathrm{Im}\{a_{-k}\}} = \boxed{\mathrm{Re}\{a_k\} - j\,\mathrm{Im}\{a_k\}}$$

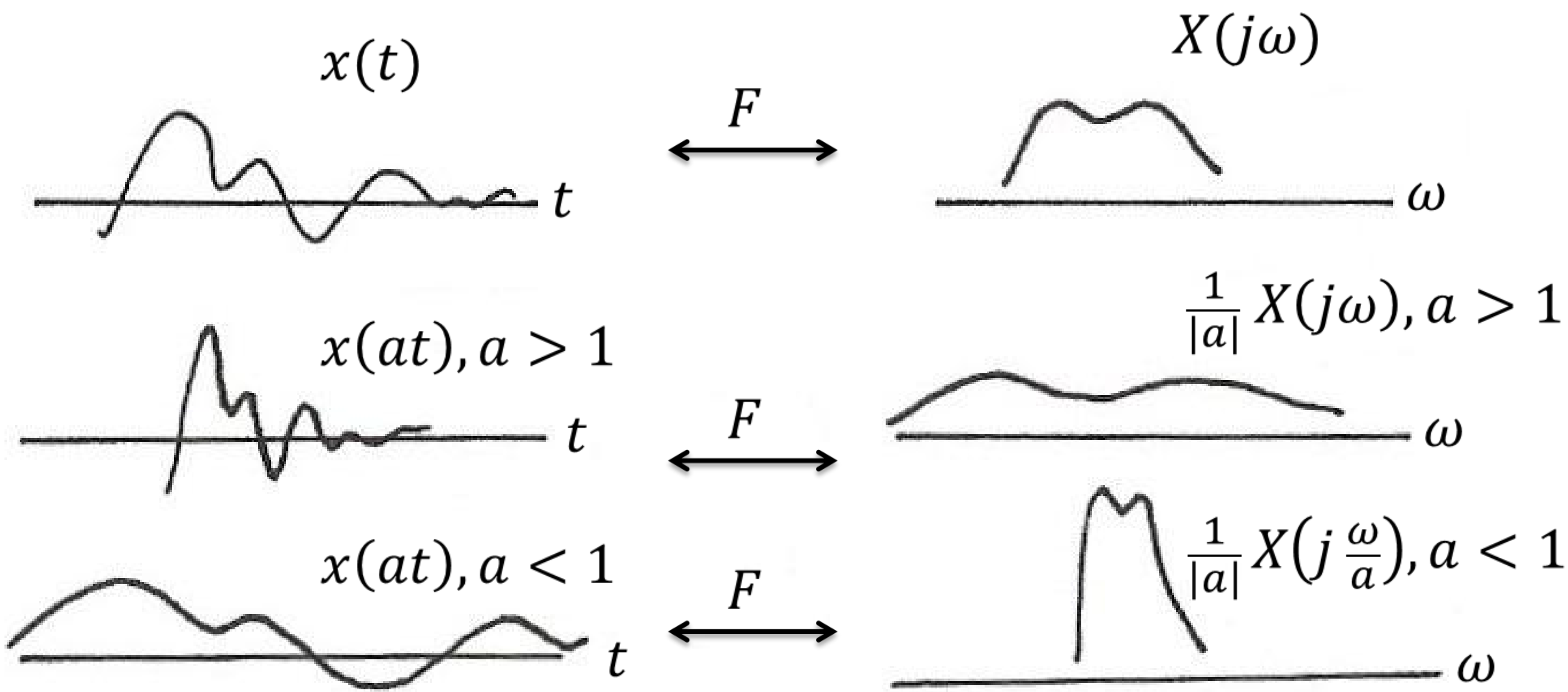
$\therefore$

$\mathrm{Re}\{a_k\}$ is even， $\mathrm{Im}\{a_k\}$ is odd

CTFT Properties (cont.)

4) Time/Frequency Scaling $x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$

E.g. $a > 1 \rightarrow at > t$
compressed in time $\leftrightarrow$
stretched in frequency



CTFT Properties (cont.)

$\boxed{\text{4) Time/Frequency Scaling}}$ $x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$

$E.g.\ a > 1 \rightarrow at > t$
compressed in time $\leftrightarrow$
stretched in frequency

$\Downarrow a = -1$

$x(-t) \longleftrightarrow X(-j\omega)$ $\boxed{\text{Time reversal}}$

$\Downarrow$

a) $x(t)$ real and even $\quad x(t) = x(-t) = x^*(t)$

$\Rightarrow X(j\omega) = X(-j\omega) = X^*(-j\omega)$ — Real & even

b) $x(t)$ real and odd $\quad x(t) = -x(-t) = x^*(t)$

$\Rightarrow X(j\omega) = -X(-j\omega) = X^*(-j\omega)$ — Purely imaginary & odd

c) $X(j\omega) = \text{Re}\{X(j\omega)\} + j\,\text{Im}\{X(j\omega)\}$

For real $x(t) = Ev\{x(t)\} + Od\{x(t)\}$

CTFT Properties (cont.)

5) Differentiation/Integration

$u(t) = 1/2sgn(t) + 1/2$
$U(jw) = 1/(jw) + \pi\delta(w)$

$$\frac{dx(t)}{dt} \xleftrightarrow{F} j\omega X(j\omega)$$

$$\int_{-\infty}^{t} x(\tau)\, d\tau \xleftrightarrow{F} \frac{1}{j\omega} X(j\omega) + \pi X(\underbrace{j0}_{\text{DC term}})\delta(\omega)$$

Example:

What is the Fourier transform for unit step function $u(t)$?

Integrators

$$x(t) \longrightarrow \boxed{\int} \longrightarrow y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$$

Impulse response: $\boxed{u_{-1}(t) \equiv u(t)}$

Operational definition: $x(t) * u_{-1}(t) = \int_{-\infty}^{t} x(\tau)\,d\tau$

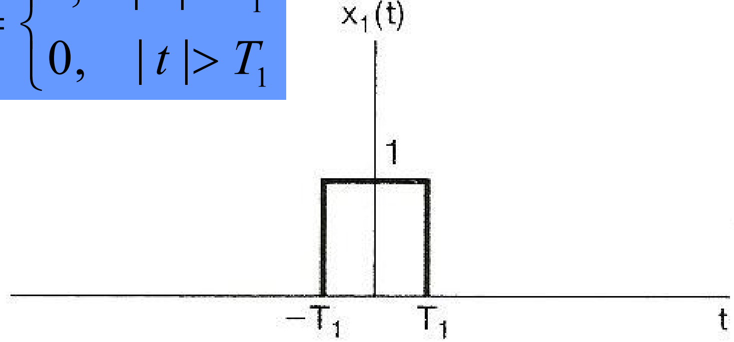
Cascade of n integrators:

$$u_{-n}(t) = \underbrace{u_{-1}(t) * \cdots * u_{-1}(t)}_{n\ times} \qquad (n > 0)$$

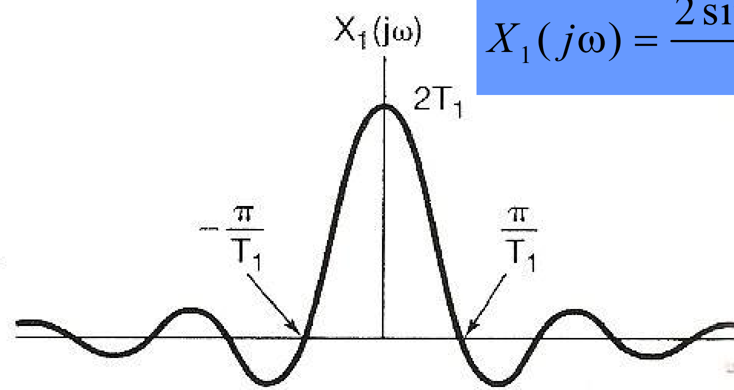
CTFT Properties (cont.)

6) Duality

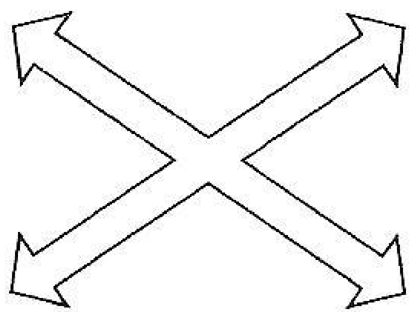
$$x_1(t)=\begin{cases}1, & |t|<T_1\\ 0, & |t|>T_1\end{cases}$$



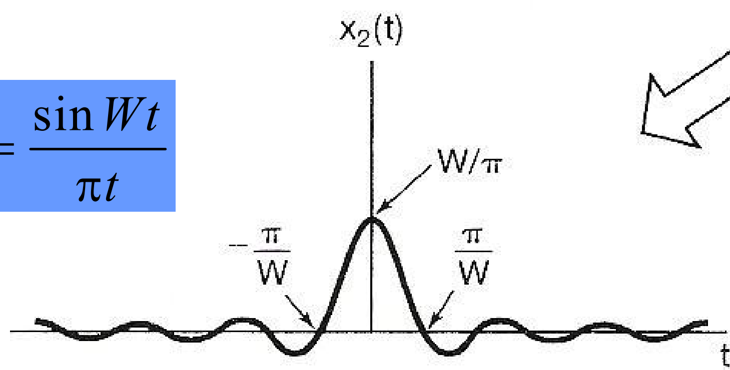
$\overset{\mathcal{F}}{\longleftrightarrow}$



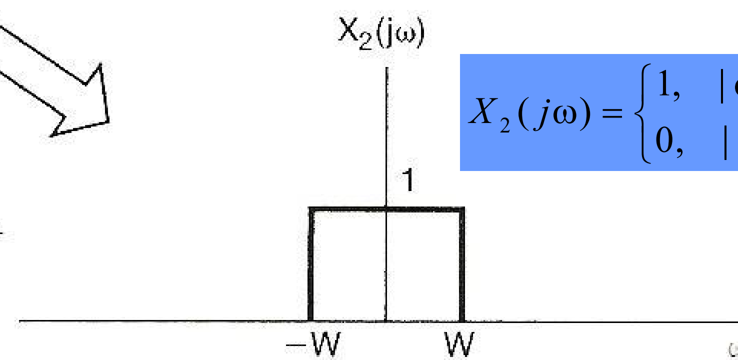
$$X_1(j\omega)=\frac{2\sin\omega T_1}{\omega}$$



$$x_2(t)=\frac{\sin Wt}{\pi t}$$



$\overset{\mathcal{F}}{\longleftrightarrow}$

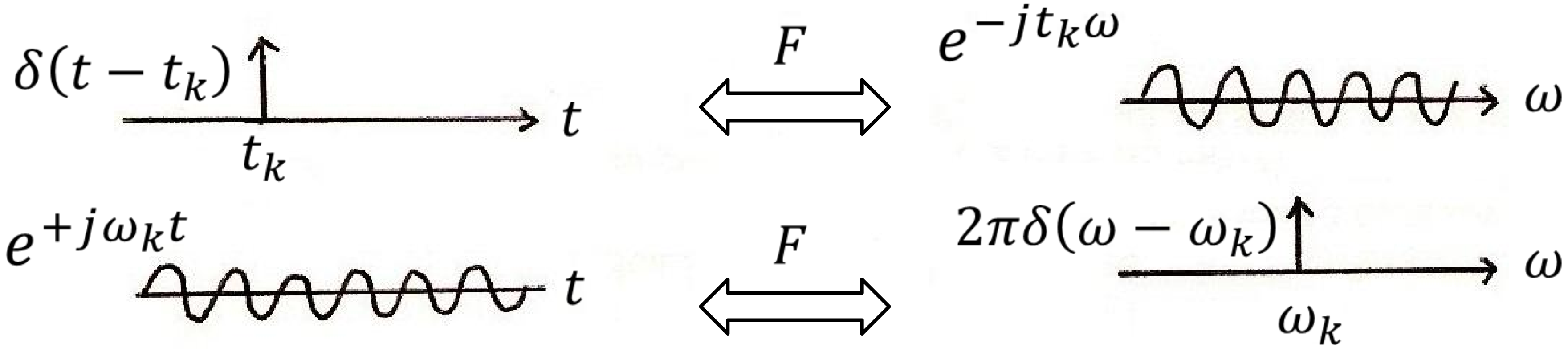


$$X_2(j\omega)=\begin{cases}1, & |\omega|<W\\ 0, & |\omega|>W\end{cases}$$

- Time/frequency domains are kind of "symmetric".
- If there are characteristics of a function of time that have implications with regard to the Fourier transform, then the same characteristics associated with a function of frequency will have ***dual*** implications in the time domain.

<u>**Example:**</u>

$\{\delta(t-t_k), -\infty < t_k < \infty\}$ $\{2\pi\delta(\omega-\omega_k), -\infty < \omega_k < \infty\}$



CTFT Properties (cont.)

7) Parseval's Relation

$$\underbrace{\int_{-\infty}^{+\infty} |x(t)|^2 \, dt}_{\text{Total energy in the time-domain}} = \underbrace{\frac{1}{2\pi}\int_{-\infty}^{+\infty} |X(j\omega)|^2 \, d\omega}_{\text{Total energy in the frequency-domain}}$$

$\frac{1}{2\pi}|X(j\omega)|^2$

— spectral density

Impulse & Frequency Responses

$$x(t)=\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longrightarrow \boxed{h(t)} \longrightarrow y(t)=\sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

$$\boxed{H(j\omega)=\int_{-\infty}^{+\infty} h(t)e^{-j\omega t}dt}$$

Impulse response $\overset{\mathbf{F}}{\longleftrightarrow}$ Frequency response

Remark:
Not every LTI system has frequency response; but according to Dirichlet conditions, most of stable LTI systems have.

Convolution Property

$$\boxed{y(t) = h(t) * x(t) \longleftrightarrow Y(j\omega) = H(j\omega)X(j\omega)}$$

An intuitive interpretation:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega \Rightarrow x(t) = \int_{-\infty}^{+\infty} \underbrace{\frac{1}{2\pi}X(j\omega)d\omega}_{coefficient}\ e^{j\omega t}$$

$$ae^{j\omega t} \longrightarrow \boxed{h(t)} \longrightarrow aH(j\omega)e^{j\omega t}$$

$$y(t) = \int_{-\infty}^{+\infty} \underbrace{H(j\omega)\frac{1}{2\pi}X(j\omega)d\omega}_{new\ coefficient}\ e^{j\omega t} \Longrightarrow y(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} H(j\omega)X(j\omega)e^{j\omega t}d\omega$$

$$Y(j\omega) = H(j\omega)\ X(j\omega)$$

LTI Systems by LCCDE

(Linear-constant-coefficient differential equations)

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$$

Using the Differentiation Property

$$\frac{d^k x(t)}{dt^k} \longleftrightarrow (j\omega)^k X(j\omega)$$

$\Downarrow$ Transform both sides of the equation

$$\sum_{k=0}^{N} a_k (j\omega)^k Y(j\omega) = \sum_{k=0}^{M} b_k (j\omega)^k X(j\omega)$$

$$\Downarrow$$

$$Y(j\omega) = \underbrace{\left[\frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k}\right]}_{H(j\omega)} X(j\omega)$$

$$\boxed{H(j\omega) = \left[\frac{\sum_{k=0}^{M} b_k (j\omega)^k}{\sum_{k=0}^{N} a_k (j\omega)^k}\right]}$$

Partial Fraction Expansion

Partial Fraction Expansion (No identical poles):

$$H(j\omega) = \frac{b_M(j\omega + z_1)(j\omega + z_2)\ldots(j\omega + z_M)}{a_N(j\omega + p_1)(j\omega + p_2)\ldots(j\omega + p_N)}$$
$$= \frac{A_1}{j\omega + p_1} + \frac{A_2}{j\omega + p_2} + \ldots + \frac{A_N}{j\omega + p_N}$$

Partial Fraction Expansion (with identical poles):

$$H(j\omega) = \frac{b_M(j\omega + z_1)(j\omega + z_2)\ldots(j\omega + z_M)}{a_N(j\omega + p_1)^{k_1}(j\omega + p_2)^{k_2}\ldots(j\omega + p_n)^{k_n}} =$$
$$= \frac{A_{1,1}}{(j\omega + p_1)^{k_1}} + \frac{A_{1,2}}{(j\omega + p_1)^{k_1 - 1}} + \ldots \frac{A_{1,k_1}}{(j\omega + p_1)}$$
$$+ \ldots +$$
$$\frac{A_{n,1}}{(j\omega + p_n)^{k_n}} + \frac{A_{n,2}}{(j\omega + p_n)^{k_n - 1}} + \ldots \frac{A_{n,k_n}}{(j\omega + p_n)}$$

DTFT

- Therefore, we get the discrete-time Fourier transform pair

Discrete-Time Fourier Transform

$$\text{Synthesis Equation:} \quad x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$$

$$\text{Analysis Equation:} \quad X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- Observations
  - Continuous spectrum: Similar to CTFT
  - Periodic with period 2π: Different from CTFT
  - Low frequency: close to 0 and 2π; high frequency: close to π

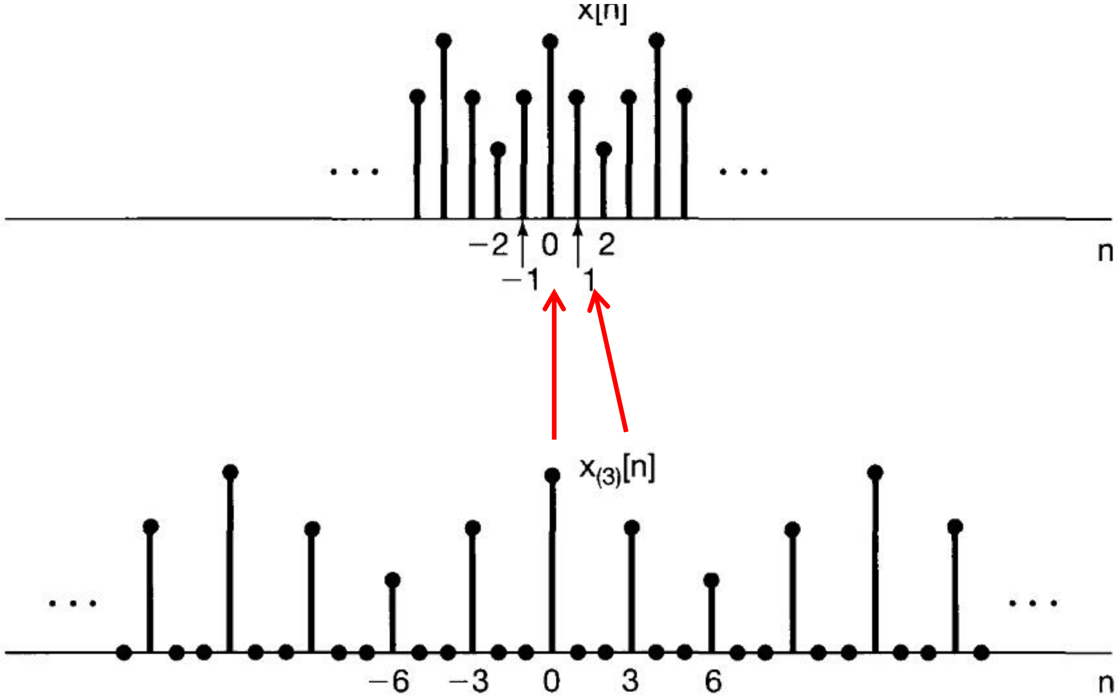
Time Reversal and Expansion

- Time Reversal

$$x[-n] \longleftrightarrow X(e^{-j\omega})$$

- Time Expansion
  - Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of k} \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$



Differentiation and Parseval

- Differentiation in Frequency

$$nx[n] \longleftrightarrow j\frac{dX(e^{j\omega})}{d\omega}$$

- Parseval's Relation

$$\sum_{n=-\infty}^{+\infty} |x[n]|^2 = \frac{1}{2\pi}\int_{2\pi} |X(e^{j\omega})|^2 d\omega$$

  - Energy density spectrum — $\frac{|X(e^{j\omega})|^2}{2\pi}$

Convolution Property & LTI Systems (1/2)

- Let $h[n]$ be the impulse response of certain LTI system
- The output $y[n]$ of input $x[n]$ is given by $y[n] = x[n] * h[n]$
- For input signal $x[n] = e^{j\omega n}$,

$$y[n] = \sum_{k=-\infty}^{\infty} h[k] e^{j\omega(n-k)} = e^{j\omega n} \underbrace{\sum_{k=-\infty}^{\infty} h[k] e^{-j\omega k}}_{\text{Frequency Response } H(e^{j\omega})}$$

- For periodic input signal $x[n] = \sum_{k=<N>} a_k e^{jk(\frac{2\pi}{N})n}$:

$$y[n] = \sum_{k=<N>} a_k H(e^{jk(\frac{2\pi}{N})}) e^{jk(\frac{2\pi}{N})n}$$

$$b_k = a_k H(e^{jk(\frac{2\pi}{N})})$$

Convolution Property & LTI Systems (2/2)

Convolution Property

If $y[n] = x_1[n] * x_2[n]$, *then*

$$Y(e^{j\omega}) = X_1(e^{j\omega})X_2(e^{j\omega})$$

- For general input signal $x[n]$:

$$y[n] = x[n] * h[n] \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$

- Observation: It's easier to evaluate LTI systems in frequency domain
- Drawback: Not every LTI system has frequency response
  - $h[n] = a^n u[n]$ $(a > 1)$
  - Stable LTI system has frequency response, because

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Linear constant-coefficient difference equations (LCCDE)

$$nx[n] \stackrel{\mathfrak{F}}{\longleftrightarrow} j\frac{dX(e^{j\omega})}{d\omega}$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})}$$

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k].$$

$$\Downarrow$$

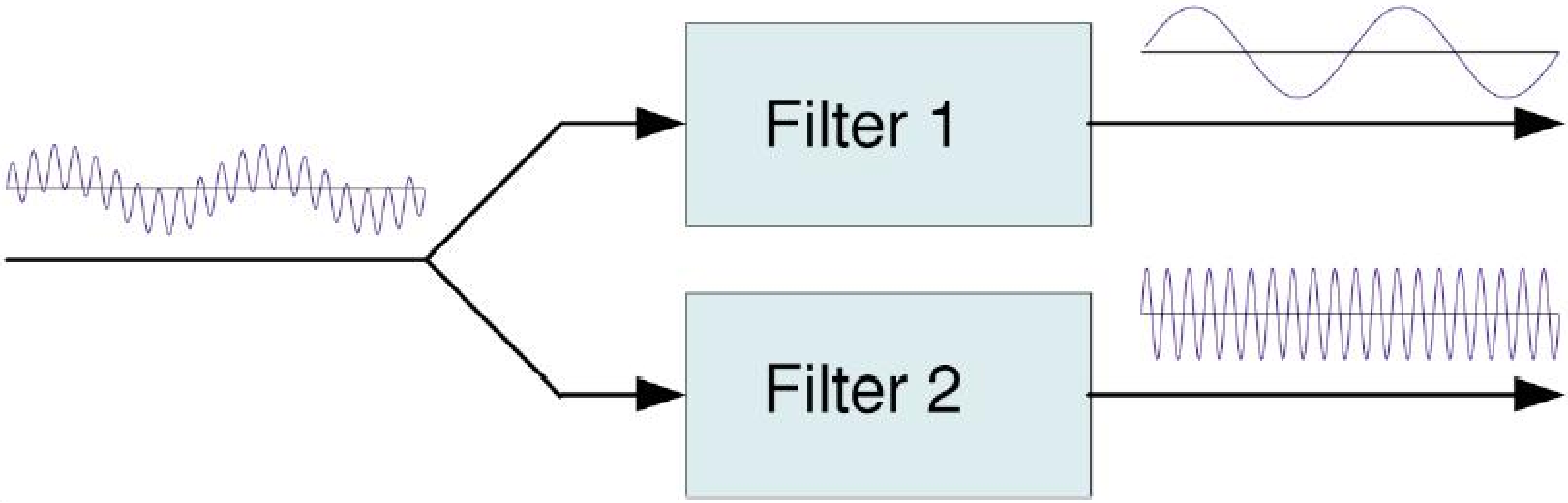
$$\sum_{k=0}^{N} a_k e^{-jk\omega} Y(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-jk\omega} X(e^{j\omega})$$

$$\Downarrow$$

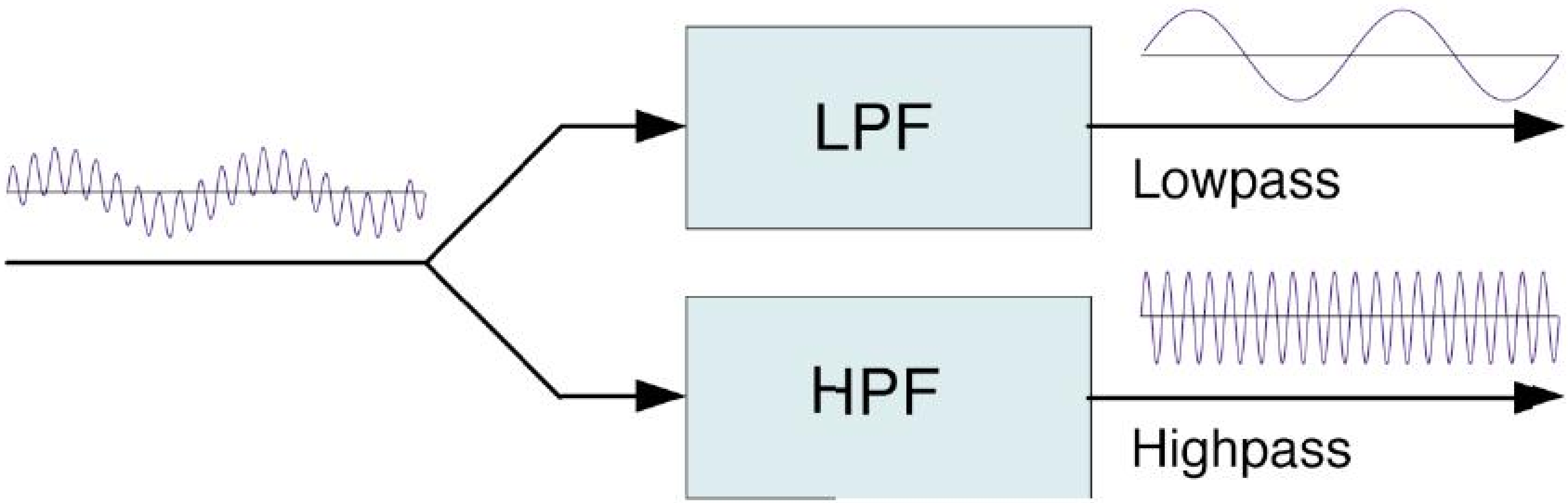
$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^{M} b_k e^{-jk\omega}}{\sum_{k=0}^{N} a_k e^{-jk\omega}}.$$

Signal Processing with Filters

- Separation of narrowband signals
  - Filter design — a main task in constructing communication systems.



Example: Using LPF and HPF to Separate Signals



Filter design becomes more challenging if the two signals are close in frequency range.

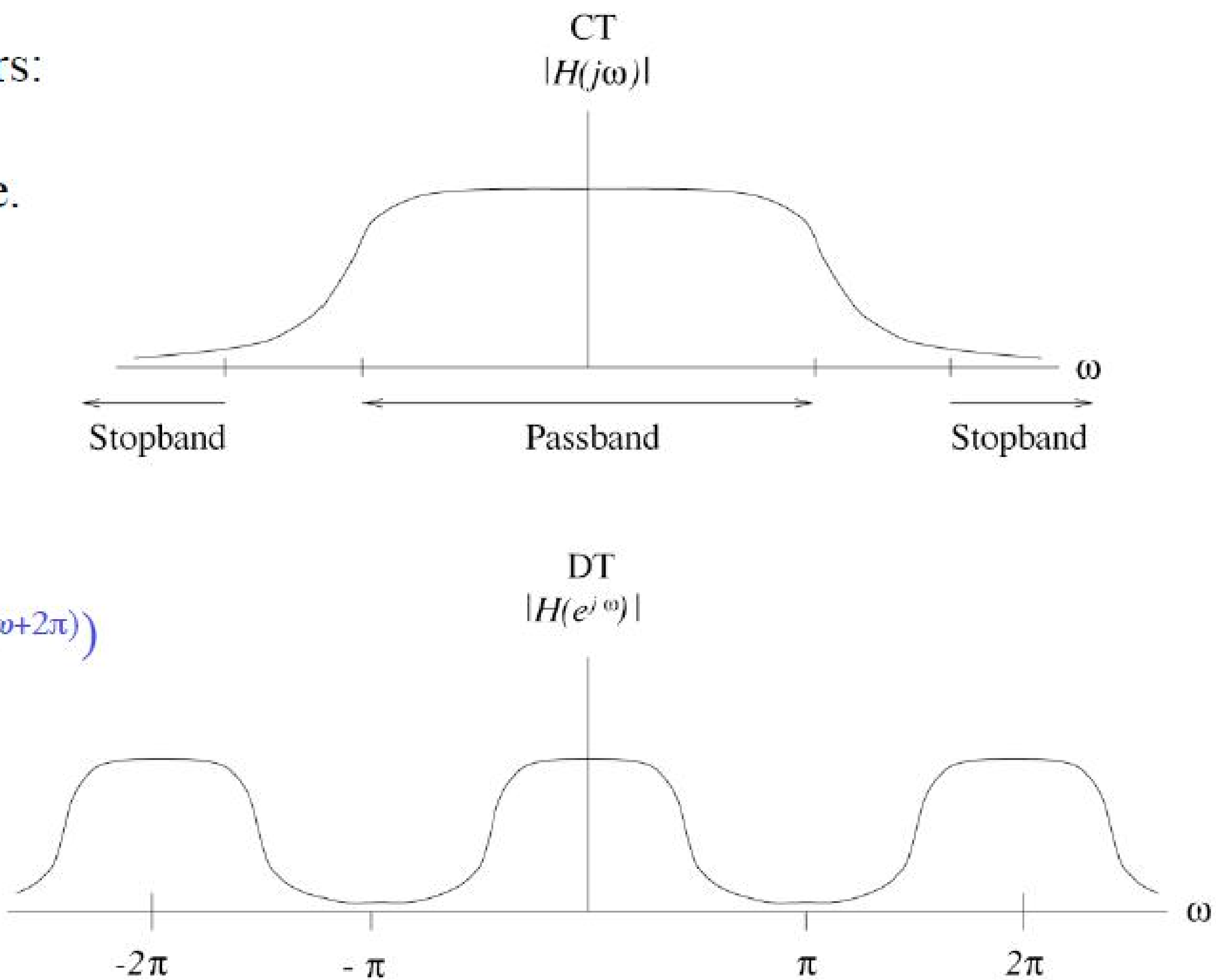
Extraction of Narrowband Signals from Wideband Noise – BPF

SUSTech Southern University of Science and Technology

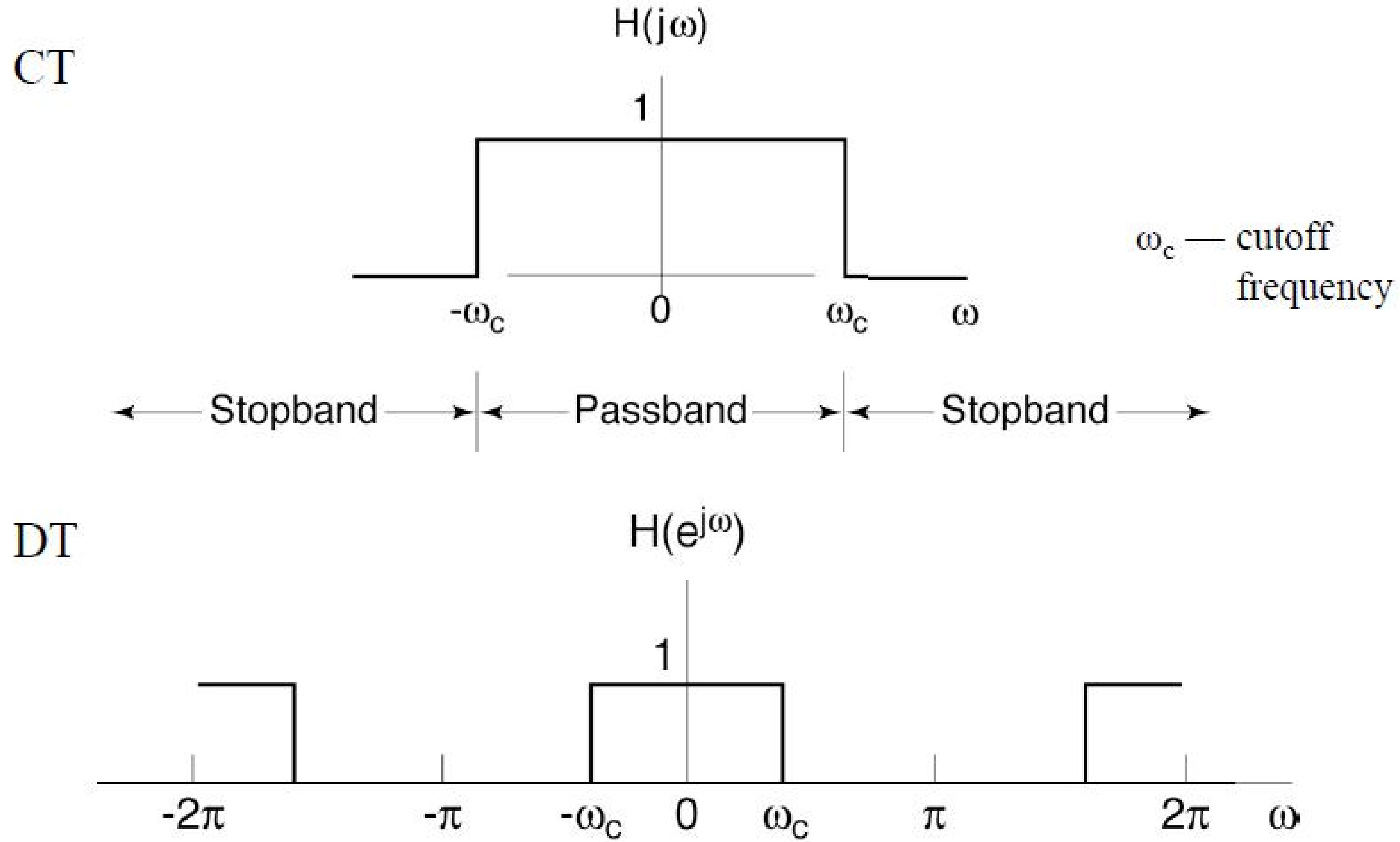
Lowpass Filter

Lowpass Filters:
Only show
amplitude here.

Note for DT:
$H(e^{j\omega}) = H(e^{j(\omega+2\pi)})$

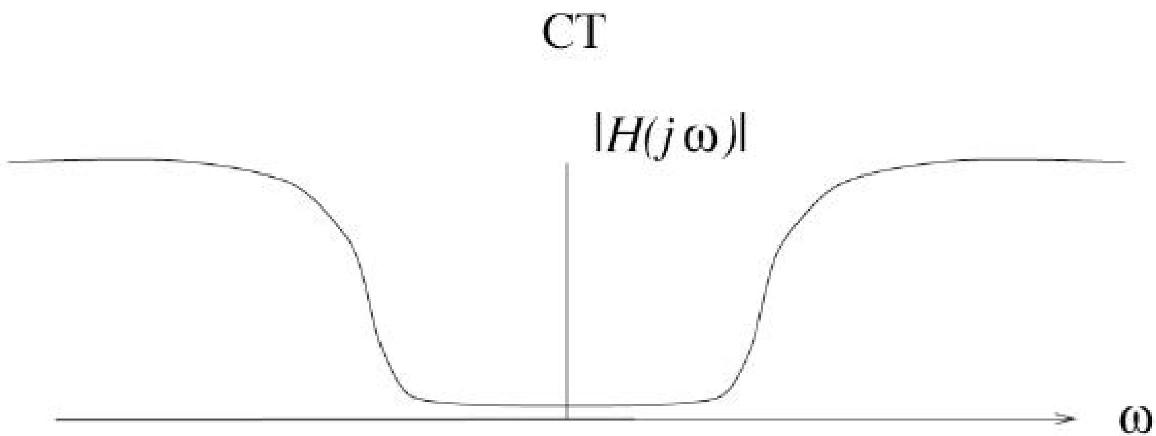


Ideal Lowpass Filter



Note: $|H| = 1$ and $\angle H = 0$ for the ideal filters in the passbands, no need for the phase plot.

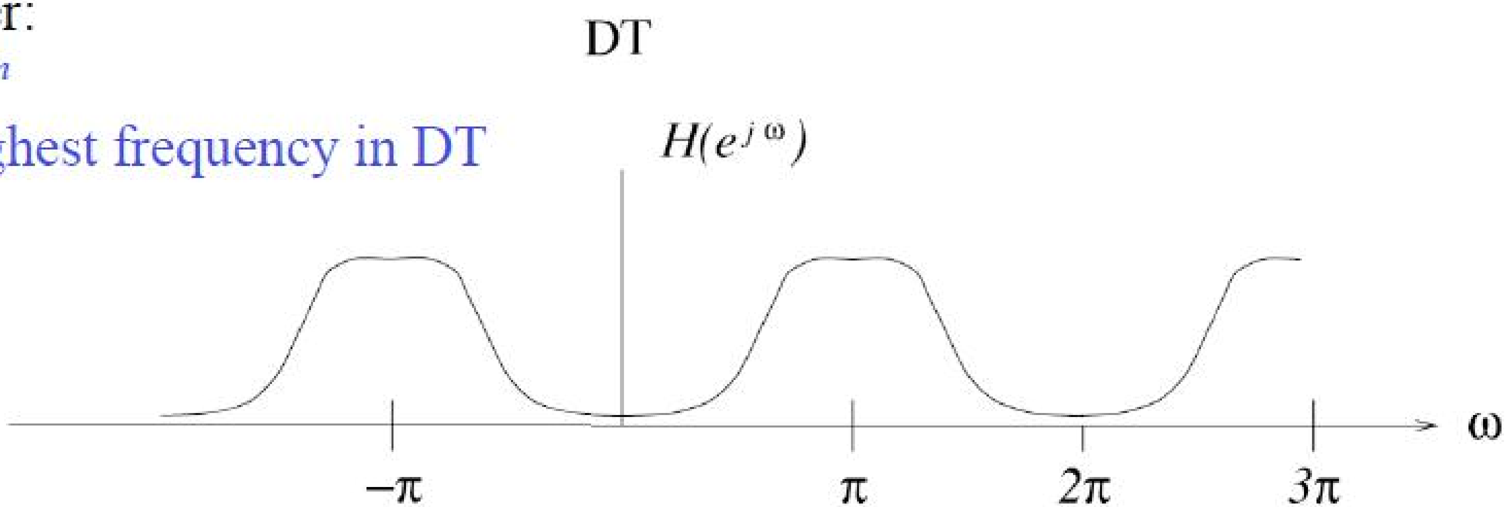
Highpass Filters



Remember:

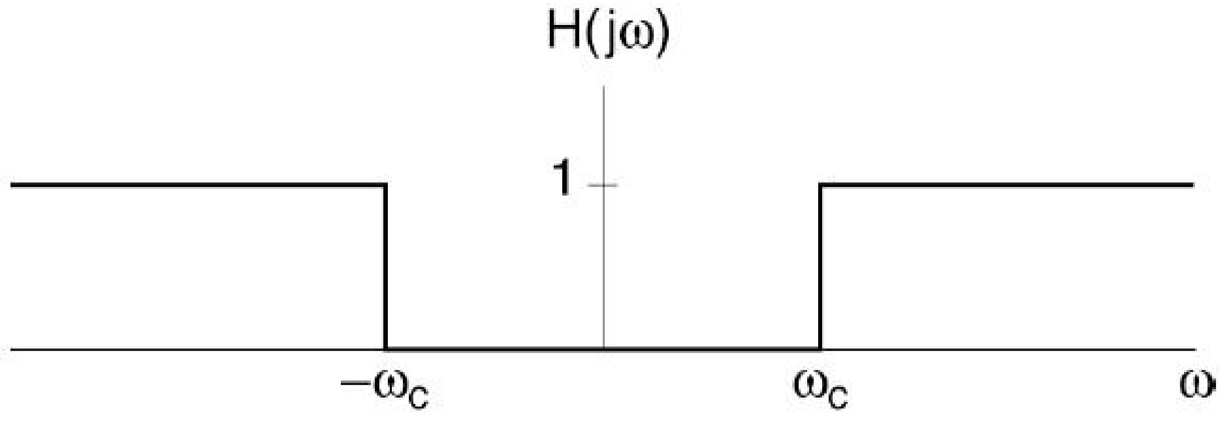
$(-1)^n = e^{j\pi n}$

— π = highest frequency in DT

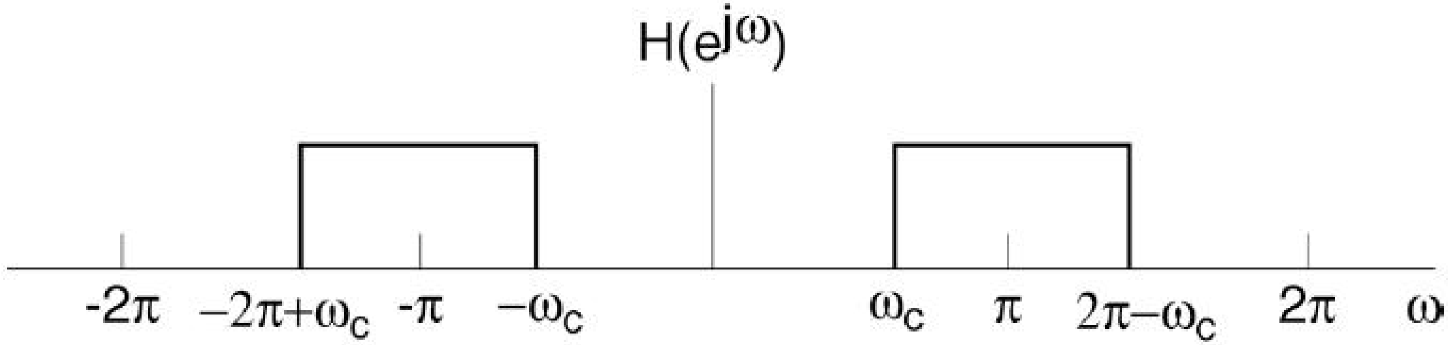


Ideal Highpass Filter

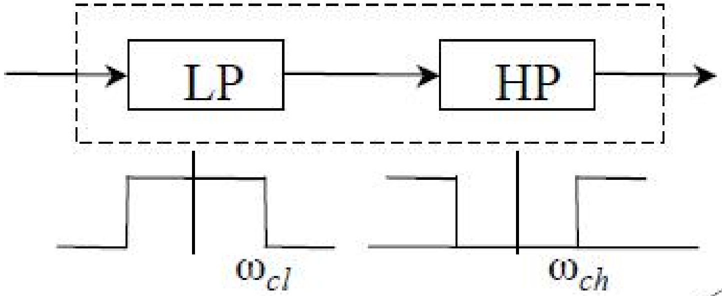
CT



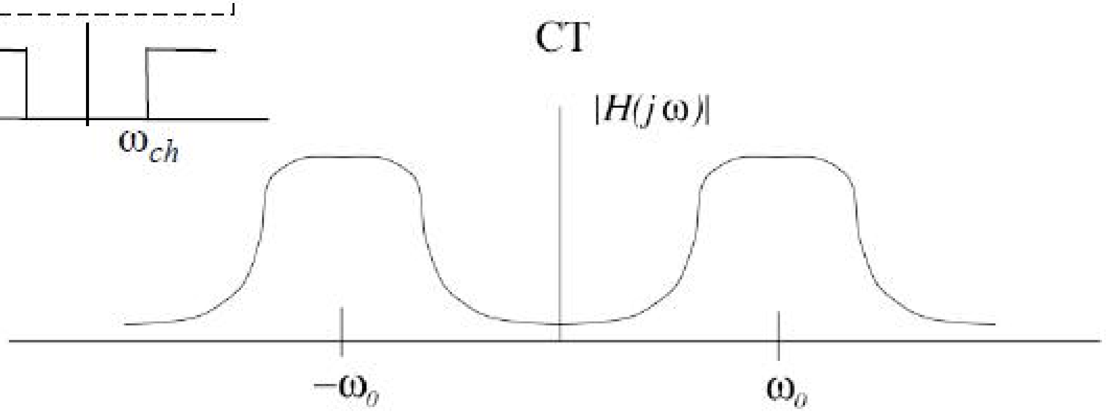
DT

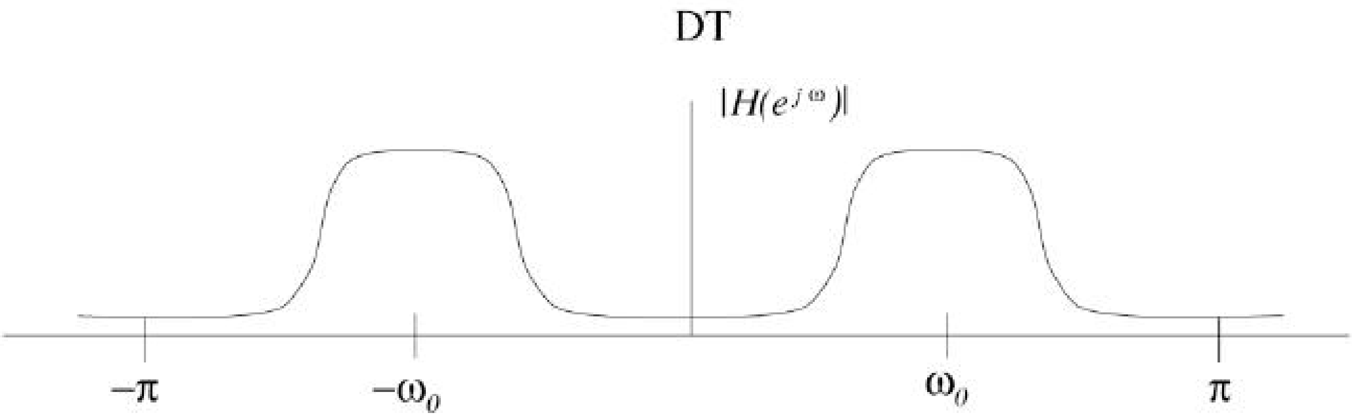


Bandpass Filters



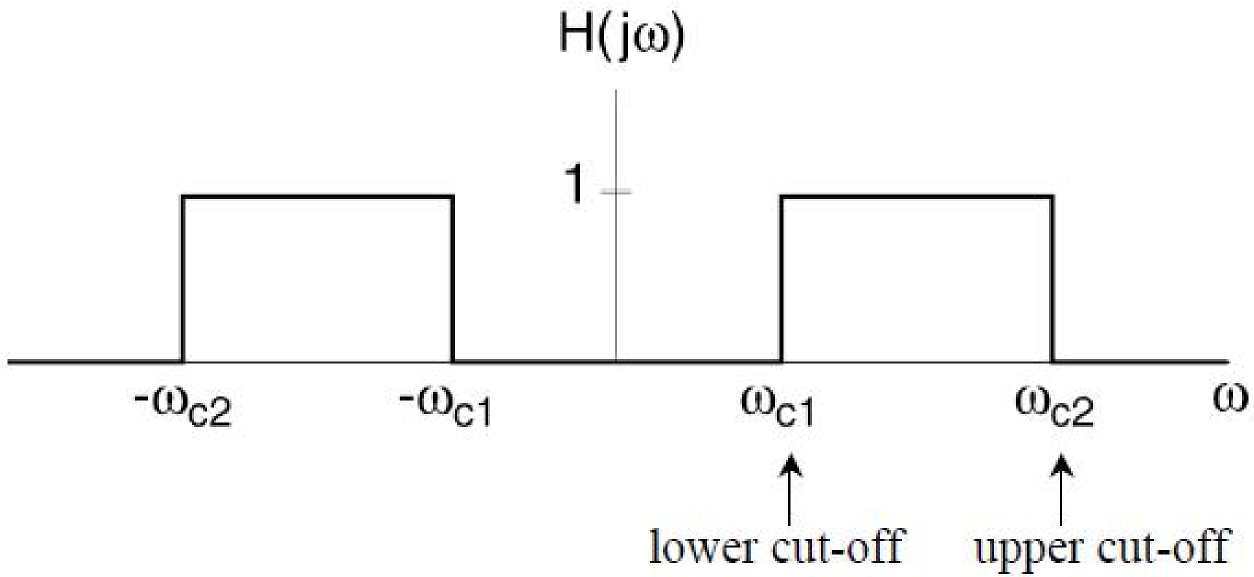
BPF if $\omega_{cl} > \omega_{ch}$



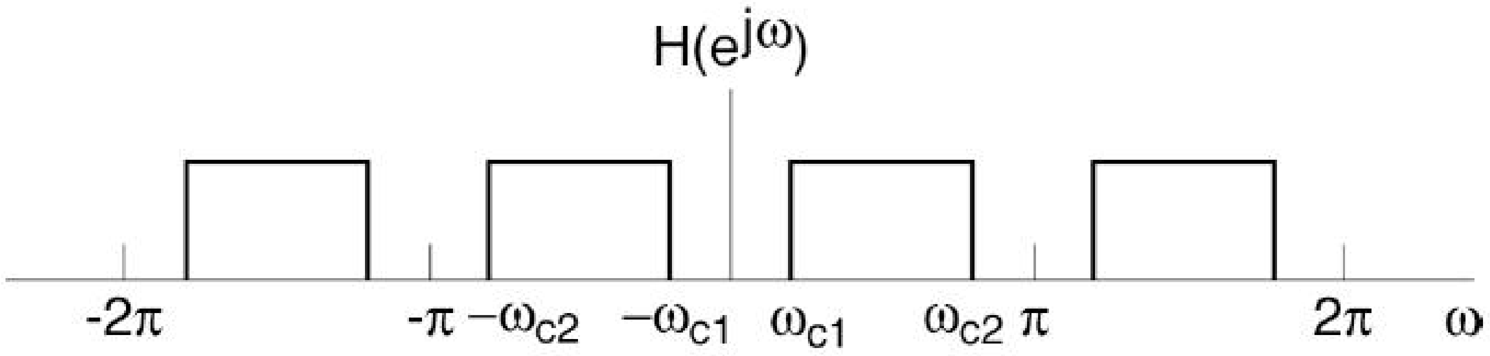


Ideal Bandpass Filter

CT

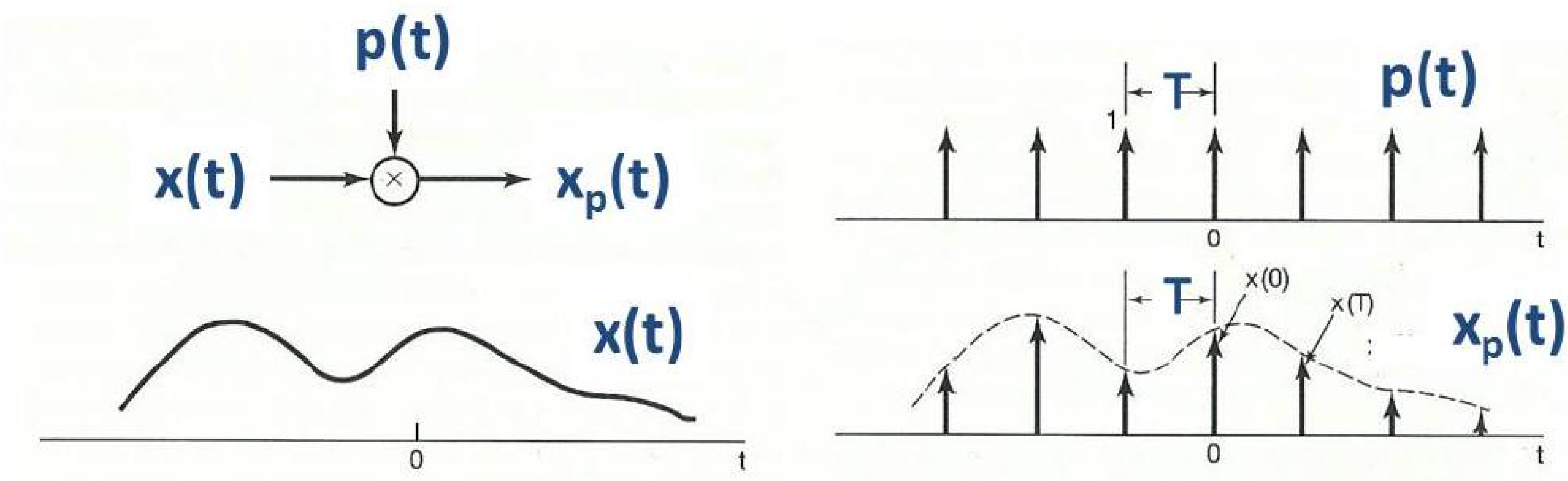


DT



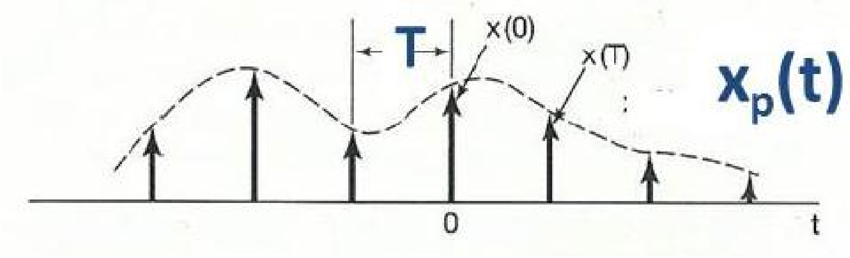
Impulse-train sampling

- Mathematically, sampling can be represented by multiplication



- Sampling function: $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$
- Sampling period: T
- Sampling:

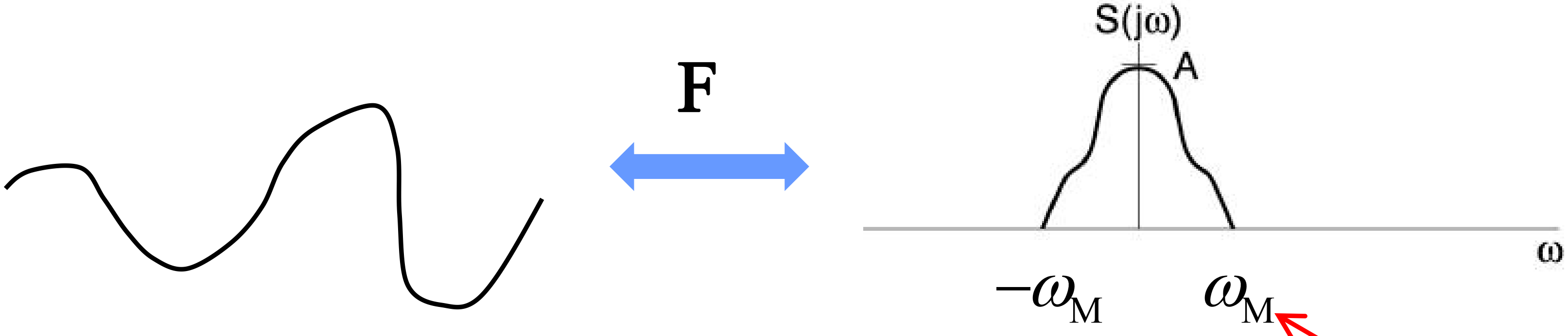
$$x_p(t) = x(t) \times p(t) = x(t) \times \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT)$$

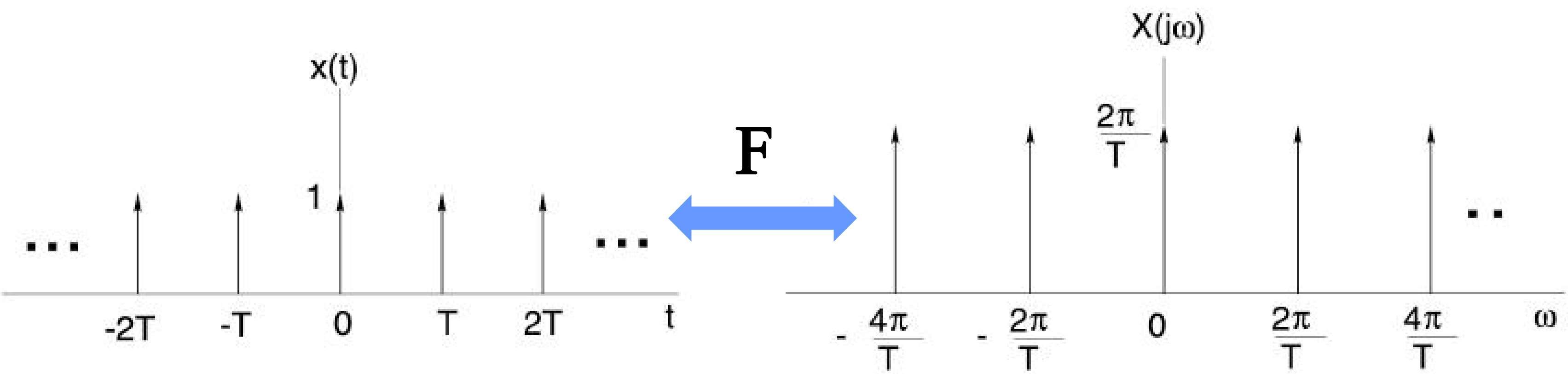


Sampling discards most of points in the original signals.

Is there any information loss in sampling?
Or can we perfectly reconstruct the original signal?

Two important frequencies





$$\omega_s = \frac{2\pi}{T}$$

Frequency analysis (1/2)

- Theoretical tool: continuous-time Fourier transform
- Principle:

$$x(t) \times p(t) \rightleftarrows \frac{1}{2\pi} X(j\omega) * P(j\omega)$$

- Fourier series of $p(t)$:

$$\begin{aligned} a_k &= \frac{1}{T} \int_{-T/2}^{T/2} \sum_{n=-\infty}^{\infty} \delta(t - nT) e^{-jk\omega_s t} dt \quad \textit{where} \quad \omega_s = \frac{2\pi}{T} \\ &= \frac{1}{T} \int_{-T/2}^{T/2} \delta(t) e^{-jk\omega_s t} dt = \frac{1}{T} \end{aligned}$$

- Fourier Transform of $p(t)$:

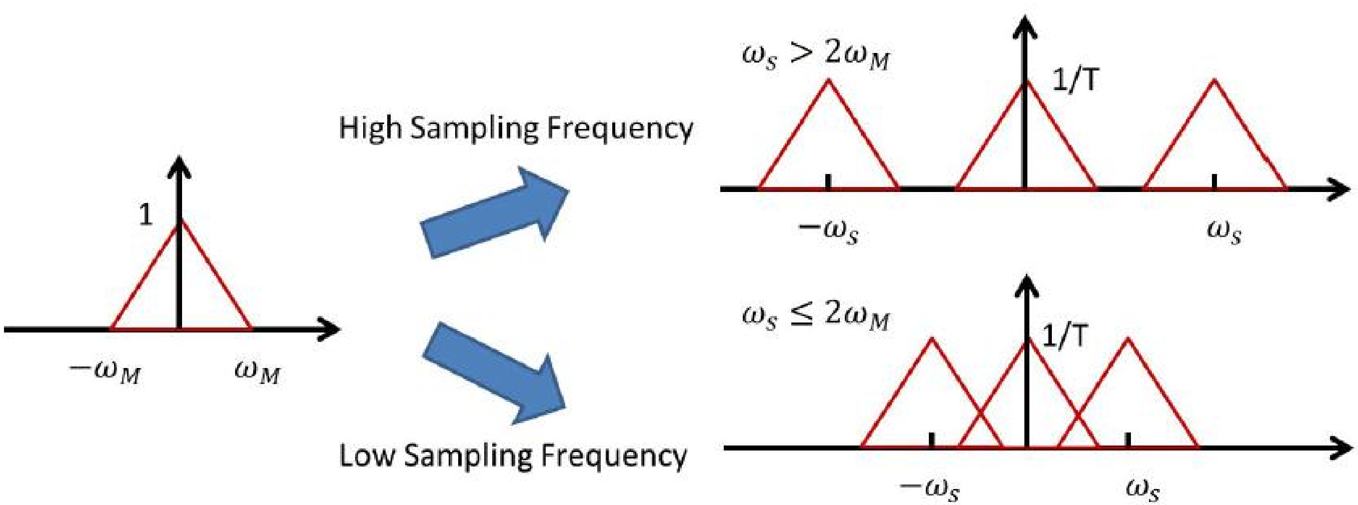
$$P(j\omega) = 2\pi a_k \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s)$$

Frequency analysis (2/2)

- Fourier transform of sampled signal $x_p(t)$:

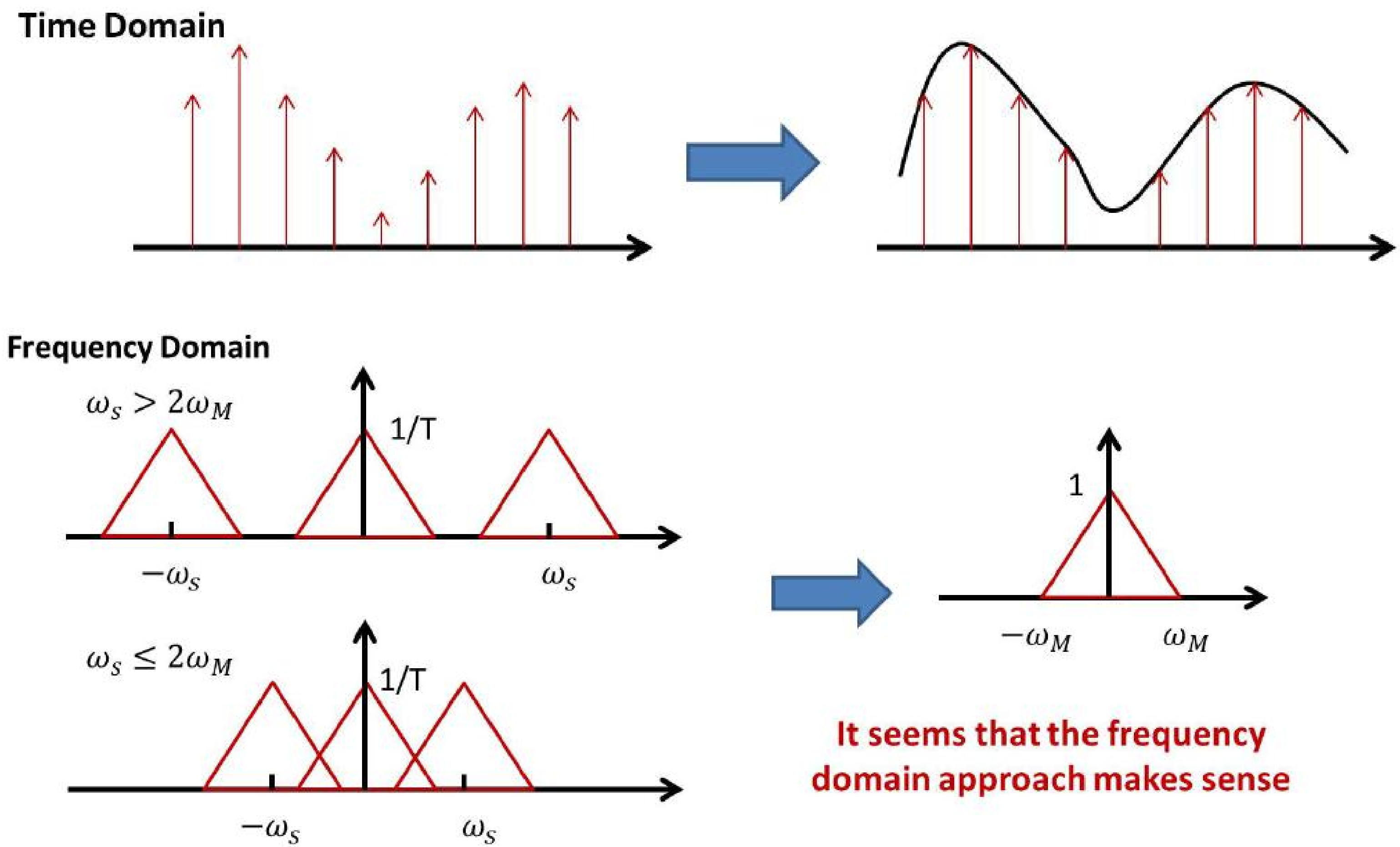
$$X_p(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)P(j(\omega-\theta))d\theta$$
$$= \frac{1}{T}\sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$$

- Sampling: the Fourier transform of input signal is repeated with period ω_s



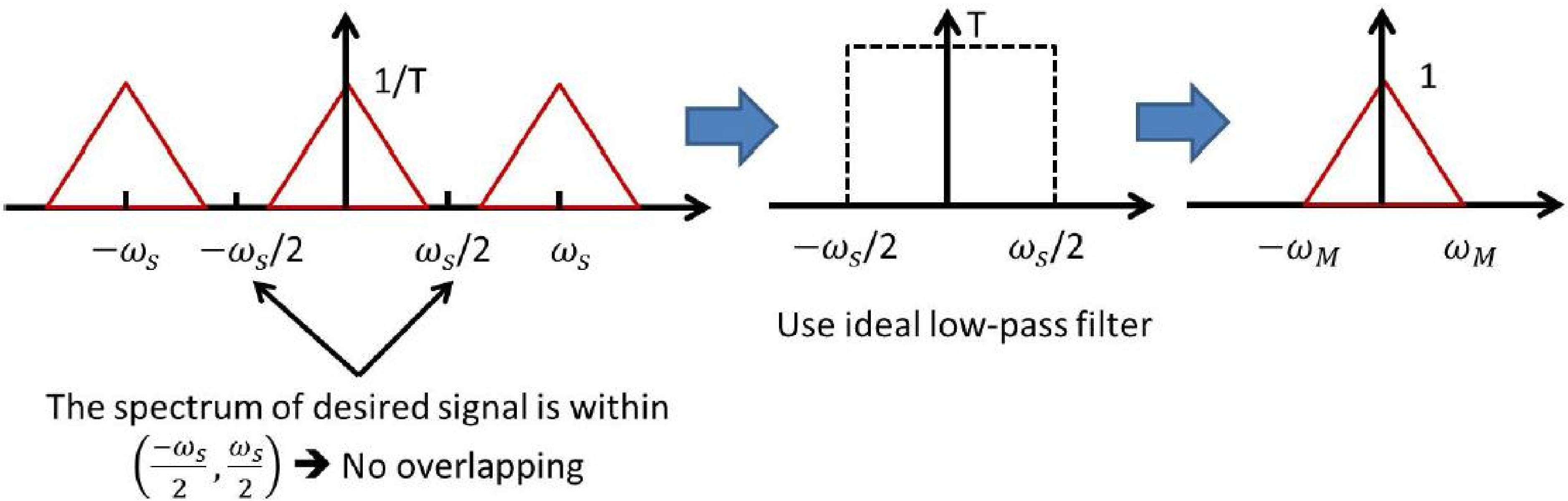
Reconstruction problem

- Given the sampled signal, can we perfectly reconstruct the signal before sampling?



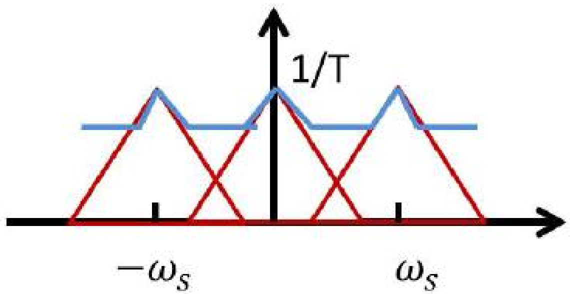
Reconstruction (1/2)

- Scenario of $\omega_s > 2\omega_M$



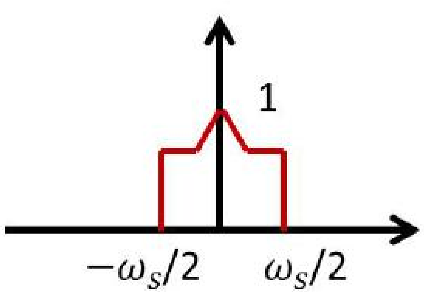
Reconstruction (2/2)

- Scenario of $\omega_s \leq 2\omega_M$

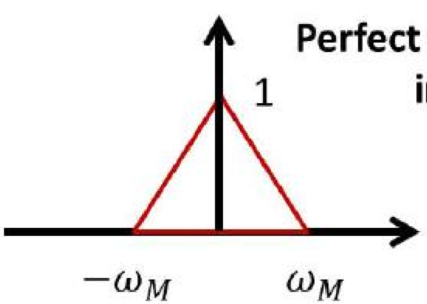


Since $\omega_s \leq 2\omega_M$, we don't know the frequency range of the desired signal

Sampling on the following signals can generate the same result:



OR



Perfect reconstruction is impossible

Observation: the original signal x(t) can be Uniquely and perfectly reconstructed from x(nT) only when $\omega_s > 2\omega_M$

Sampling theorem

Sampling Theorem

Let $x(t)$ be a band-limited signal with

$$X(j\omega) = 0 \text{ for } |\omega| > \omega_M.$$

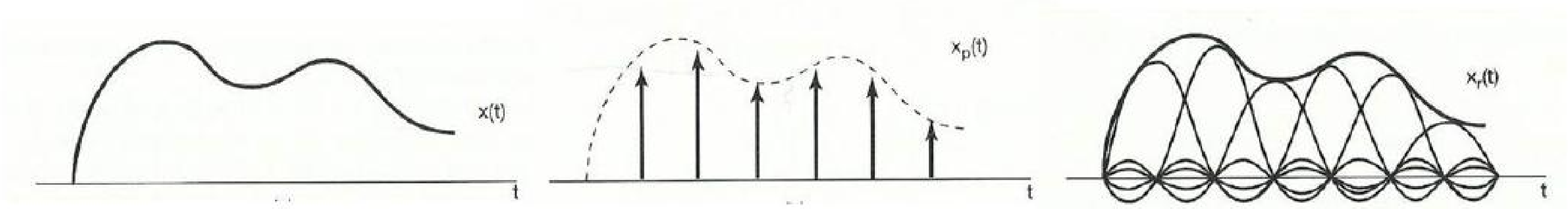
Then, $x(t)$ is uniquely determined by its samples $x(nT)$ or $x_p(t)$ if

$$\omega_s = \frac{2\pi}{T} > 2\omega_M,$$

where $2\omega_M$ is referred to as the Nyquist rate.

Signal reconstruction: Interpolation

- If $\omega_s > 2\omega_M$, original signal can be perfectly reconstructed by ideal low-pass filter.
- Time domain interpretation of lowpass filtering



$$
\begin{aligned}
x_r(t) &= x_p(t) * h(t) = \sum_{n=-\infty}^{+\infty} x(nT)h(t-nT) \\
&= \sum_{n=-\infty}^{+\infty} x(nT)\frac{\sin\frac{\omega_s}{2}(t-nT)}{\frac{\omega_s}{2}(t-nT)} = \sum_{n=-\infty}^{+\infty} x(nT)sinc(\frac{t-nT}{T})
\end{aligned}
$$

- Ideal lowpass filtering: interpolation with sinc function

Amplitude modulation with a complex exponential carrier

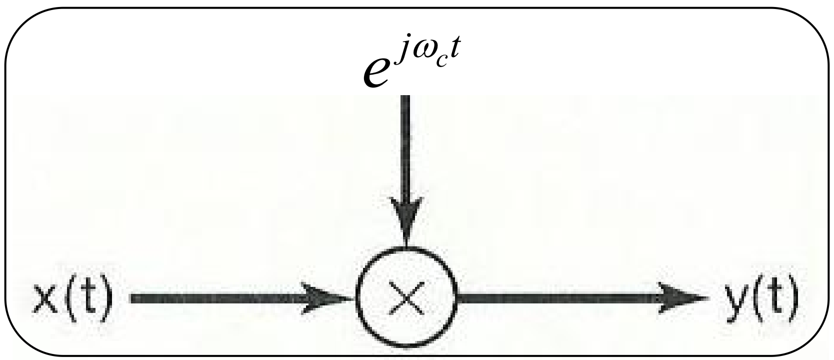
ω_M : the highest frequency in x(t)

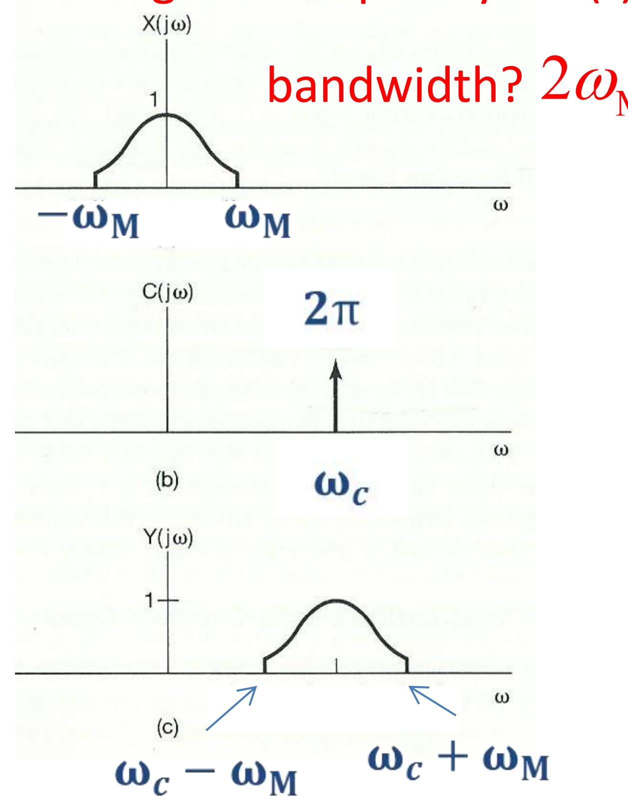
$$c(t) = e^{j(\omega_c t + \theta_c)}$$

ω_c : carrier frequency

First, we suppose $\theta_c = 0$

$$y(t) = x(t)c(t) = x(t)e^{j\omega_c t}$$

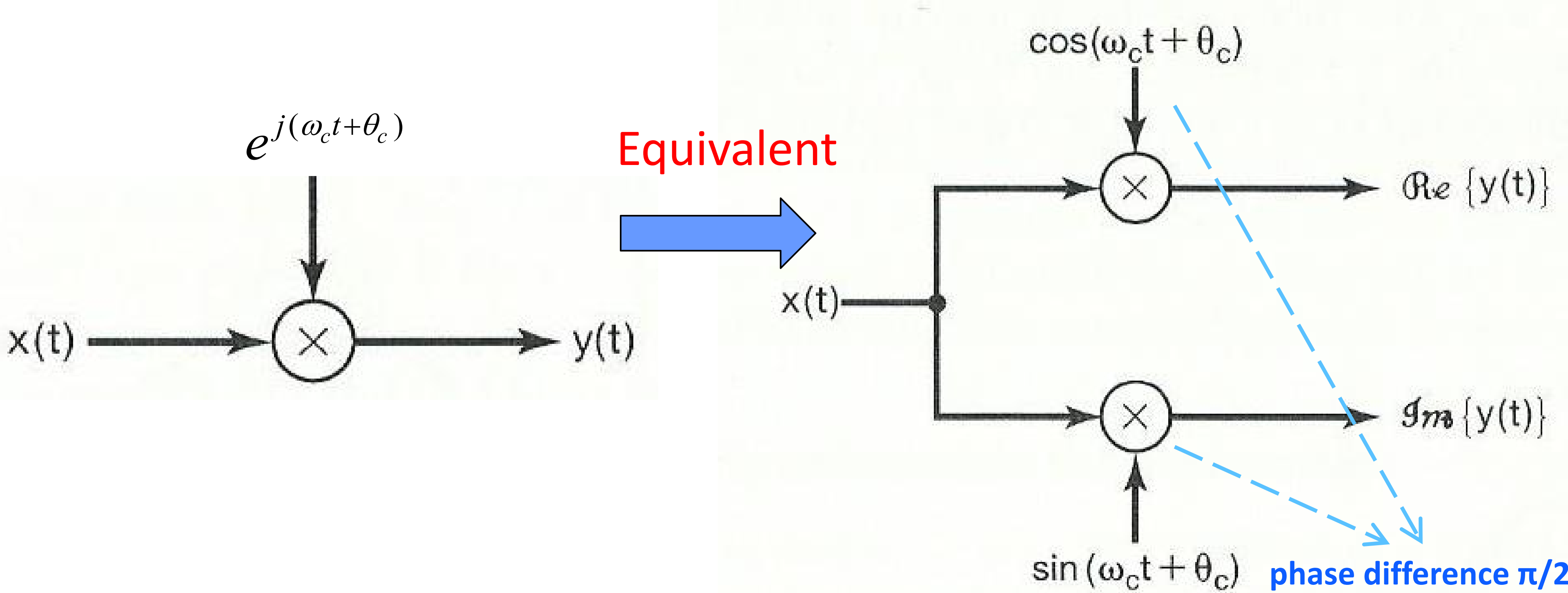




- The spectrum of the modulated output y(t) is that of modulating input x(t), shifted in frequency by amount of carrier frequency ω_c.
- How to recover the modulating input x(t) from modulated signal y(t)?

Amplitude modulation with a complex exponential carrier (cont.)

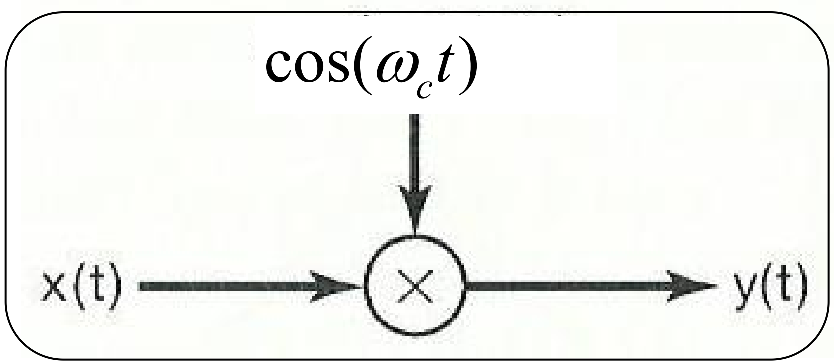
$$y(t) = x(t)e^{j\omega_c t} = x(t)\cos(\omega_c t) + jx(t)\sin(\omega_c t)$$

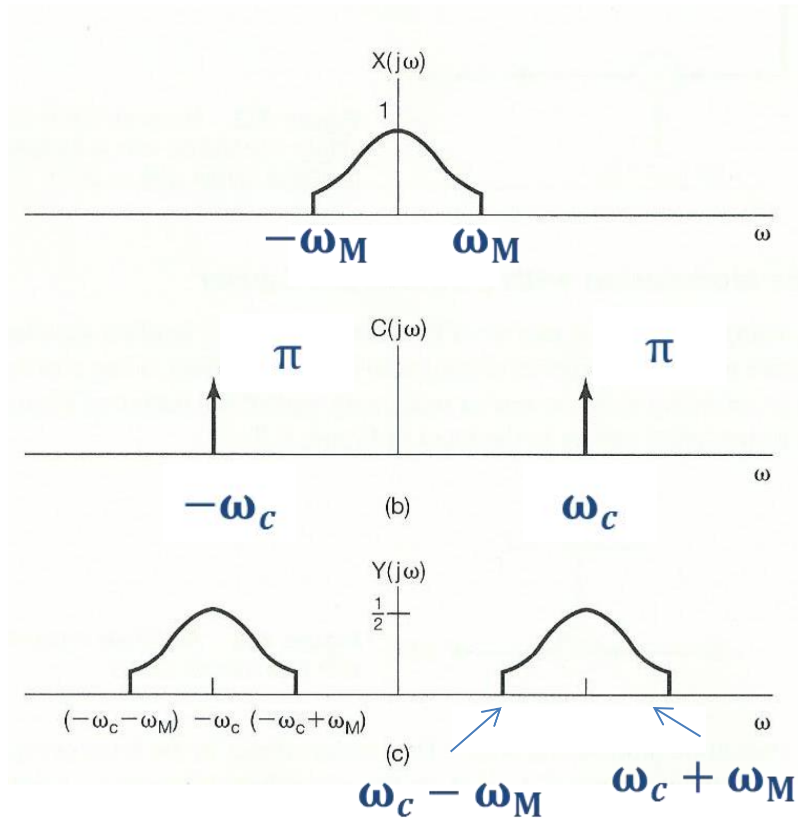


Amplitude modulation with a sinusoidal carrier

First, we suppose $\theta_c = 0$

$$y(t) = x(t)c(t) = x(t)\cos(\omega_c t)$$



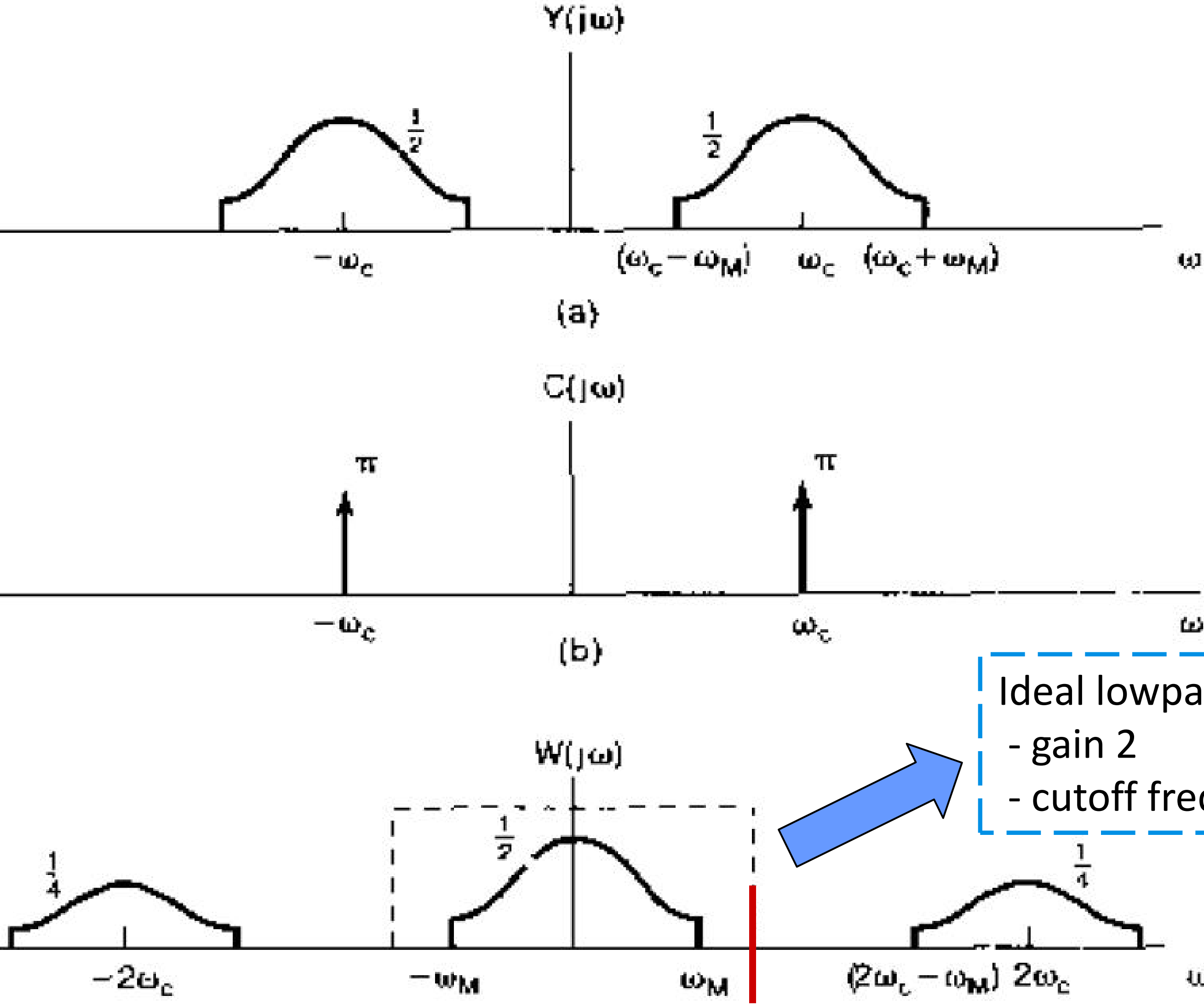


What is required if x(t) is recoverable from y(t)?

Condition: $\omega_c > \omega_M$

Synchronous demodulation

With sinusoidal carrier



$$y(t) = x(t)\cos(\omega_c t)$$

$$w(t) = y(t)\cos(\omega_c t)$$

$w(t)$: demodulated signal

Ideal lowpass filtering
- gain 2
- cutoff frequency > ω_M and < $2\omega_c - \omega_M$

Synchronous demodulation (cont.)

- **Mathematically,**

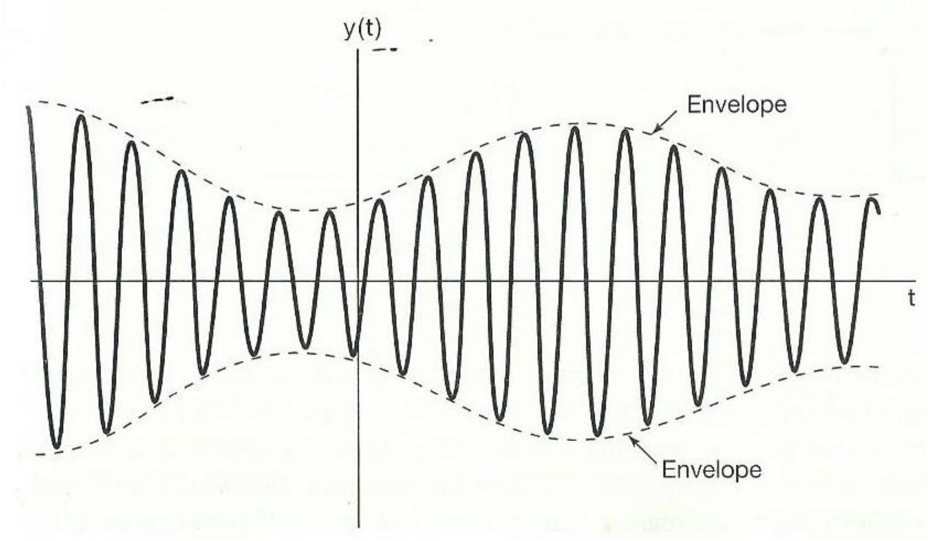
AM modulation → $$y(t) = x(t)\cos(\omega_c t)$$

AM demodulation → $$\begin{aligned} w(t) &= y(t)\cos(\omega_c t) = x(t)\cos^2(\omega_c t) \\ &= x(t)[\frac{1}{2} + \frac{1}{2}\cos(2\omega_c t)] \\ &\boxed{= \frac{1}{2}x(t)} + \frac{1}{2}x(t)\cos(2\omega_c t) \end{aligned}$$

Asynchronous demodulation

- **Avoid the need for synchronization between modulator and demodulator.**

An example of modulated signal y(t), when x(t) is positive, and $\omega_c > \omega_M$.

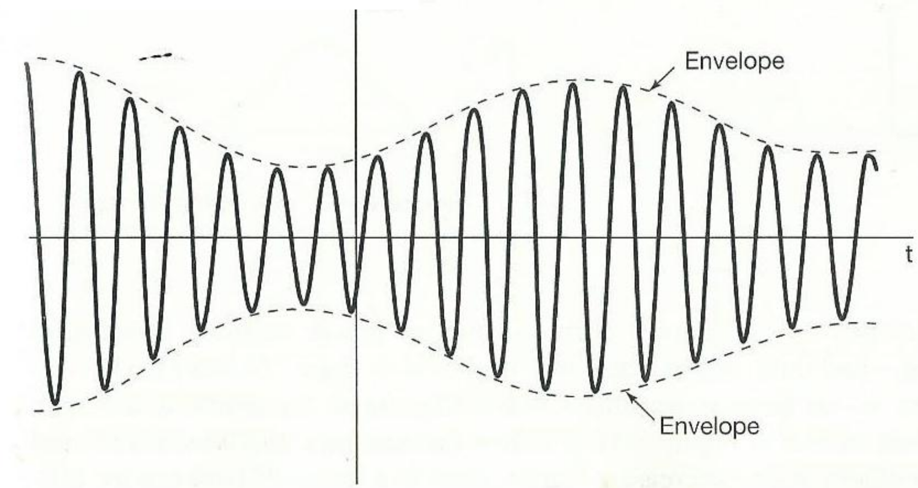


Envelope signal, a smooth curve connecting the peaks, approaches to the modulating signal x(t).

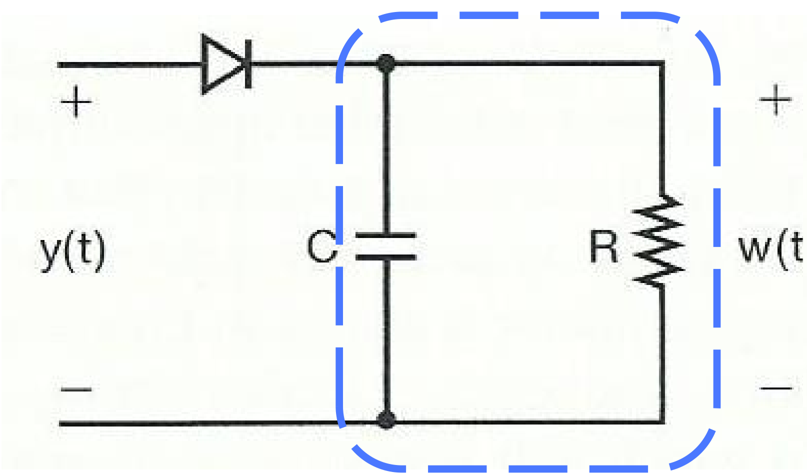
Envelope detector

Full wave or half-wave envelope detector?

Modulated signal



Lowpass filtering



Envelope detector output